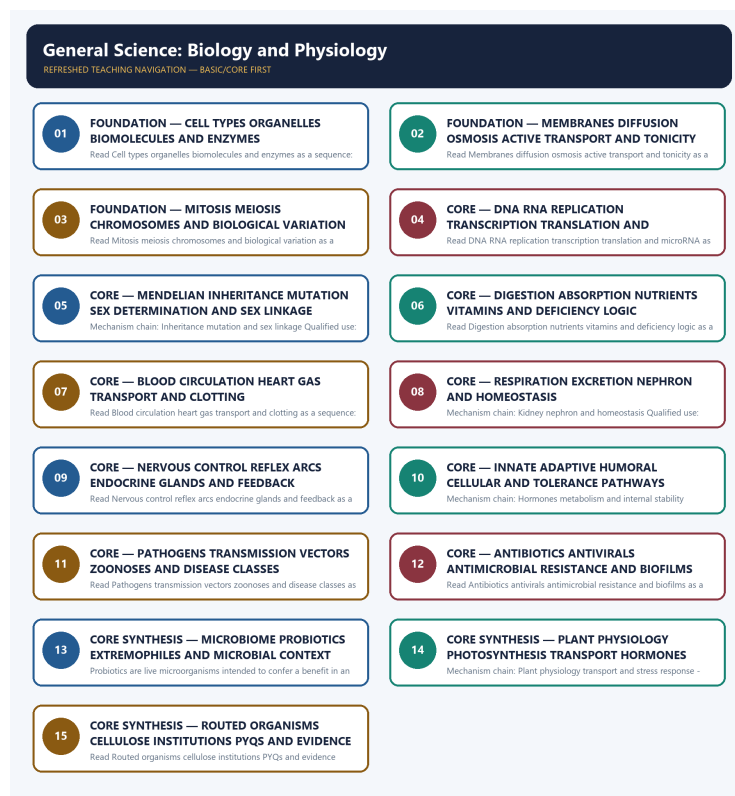


COMPLETE LEARNING SESSION

General Science: Biology and Physiology - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / SESSION INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

General Science: Biology and Physiology - Learner-v2 Complete Learning Session	6
SOURCE, PROGRESSION AND CURRENT-LINKAGE AUDIT	6
LIVE OFFICIAL-SOURCE ATTEMPT LOG	6
BASIC LEARNING SESSION	7
DEEP-REVIEW LEARNING CONTRACT	7
EVIDENCE, PYQ AND CURRENT-STATUS CONTROL	7
SESSION 1 - FOUNDATION - CELL TYPES ORGANELLES BIOMOLECULES AND ENZYMES	9
SESSION 2 - FOUNDATION - MEMBRANES DIFFUSION OSMOSIS ACTIVE TRANSPORT AND TONICITY	11
SESSION 3 - FOUNDATION - MITOSIS MEIOSIS CHROMOSOMES AND BIOLOGICAL VARIATION	12
SESSION 4 - CORE - DNA RNA REPLICATION TRANSCRIPTION TRANSLATION AND MICRORNA	14
SESSION 5 - CORE - MENDELIAN INHERITANCE MUTATION SEX DETERMINATION AND SEX LINKAGE	16
SESSION 6 - CORE - DIGESTION ABSORPTION NUTRIENTS VITAMINS AND DEFICIENCY LOGIC	18
SESSION 7 - CORE - BLOOD CIRCULATION HEART GAS TRANSPORT AND CLOTTING	20
SESSION 8 - CORE - RESPIRATION EXCRETION NEPHRON AND HOMEOSTASIS	22
SESSION 9 - CORE - NERVOUS CONTROL REFLEX ARCS ENDOCRINE GLANDS AND FEEDBACK	24
SESSION 10 - CORE - INNATE ADAPTIVE HUMORAL CELLULAR AND TOLERANCE PATHWAYS	25
SESSION 11 - CORE - PATHOGENS TRANSMISSION VECTORS ZOONOSSES AND DISEASE CLASSES	27
SESSION 12 - CORE - ANTIBIOTICS ANTIVIRALS ANTIMICROBIAL RESISTANCE AND BIOFILMS	29
SESSION 13 - CORE SYNTHESIS - MICROBIOME PROBIOTICS EXTREMOPHILES AND MICROBIAL CONTEXT	31
SESSION 14 - CORE SYNTHESIS - PLANT PHYSIOLOGY PHOTOSYNTHESIS TRANSPORT HORMONES AND ADAPTATION	33
SESSION 15 - CORE SYNTHESIS - ROUTED ORGANISMS CELLULOSE INSTITUTIONS PYQS AND EVIDENCE STATUS	35
CLOSING SEMANTIC LEDGER A - WATER, GRANT FORMS, INTERMEDIARIES AND STATE DEBATE	43
CLOSING SEMANTIC LEDGER B - NEGOTIATED CULTURE, LANGUAGE AND BOUNDARY CONTROL	44
SCIENCE AND TECHNOLOGY DEEP-REVIEW CORE CONTROL	45
BASIC MCQS / REMEDIATION	45
Q1. Which statement correctly identifies Cell organisation and organelle boundary?	45
Q2. Which option preserves the technical boundary of Cell organisation and organelle boundary?	46
Q3. Which statement uses Cell organisation and organelle boundary without changing its institution, unit or status?	46

Q4. Which option avoids the standard UPSC close-option trap about Cell organisation and organelle boundary?	47
Q5. Which statement correctly identifies Biomolecules and enzyme specificity?	47
Q6. Which option preserves the technical boundary of Biomolecules and enzyme specificity?	48
Q7. Which statement uses Biomolecules and enzyme specificity without changing its institution, unit or status?	48
Q8. Which option avoids the standard UPSC close-option trap about Biomolecules and enzyme specificity?	49
Q9. Which statement correctly identifies Membrane transport and water balance?	49
Q10. Which option preserves the technical boundary of Membrane transport and water balance?	50
Q11. Which statement uses Membrane transport and water balance without changing its institution, unit or status?	50
Q12. Which option avoids the standard UPSC close-option trap about Membrane transport and water balance?	51
Q13. Which statement correctly identifies Mitosis meiosis and variation?	51
Q14. Which option preserves the technical boundary of Mitosis meiosis and variation?	52
Q15. Which statement uses Mitosis meiosis and variation without changing its institution, unit or status?	52
Q16. Which option avoids the standard UPSC close-option trap about Mitosis meiosis and variation?	53
Q17. Which statement correctly identifies DNA RNA gene expression and microRNA?	53
Q18. Which option preserves the technical boundary of DNA RNA gene expression and microRNA?	54
Q19. Which statement uses DNA RNA gene expression and microRNA without changing its institution, unit or status?	54
Q20. Which option avoids the standard UPSC close-option trap about DNA RNA gene expression and microRNA?	55
Q21. Which statement correctly identifies Inheritance mutation and sex linkage?	55
Q22. Which option preserves the technical boundary of Inheritance mutation and sex linkage?	56
Q23. Which statement uses Inheritance mutation and sex linkage without changing its institution, unit or status?	56
Q24. Which option avoids the standard UPSC close-option trap about Inheritance mutation and sex linkage?	57
Q25. Which statement correctly identifies Digestion absorption and nutrition?	57
Q26. Which option preserves the technical boundary of Digestion absorption and nutrition?	58
Q27. Which statement uses Digestion absorption and nutrition without changing its institution, unit or status?	58
Q28. Which option avoids the standard UPSC close-option trap about Digestion absorption and nutrition?	59
Q29. Which statement correctly identifies Blood heart and double circulation?	59
Q30. Which option preserves the technical boundary of Blood heart and double circulation?	60
Q31. Which statement uses Blood heart and double circulation without changing its institution, unit or status?	60
Q32. Which option avoids the standard UPSC close-option trap about Blood heart and double circulation?	61
Q33. Which statement correctly identifies Respiration gas exchange and cellular energy?	61
Q34. Which option preserves the technical boundary of Respiration gas exchange and cellular energy?	62
Q35. Which statement uses Respiration gas exchange and cellular energy without changing its institution, unit or status?	62

Q36. Which option avoids the standard UPSC close-option trap about Respiration gas exchange and cellular energy?	63
Q37. Which statement correctly identifies Kidney nephron and homeostasis?	63
Q38. Which option preserves the technical boundary of Kidney nephron and homeostasis?	63
Q39. Which statement uses Kidney nephron and homeostasis without changing its institution, unit or status?	64
Q40. Which option avoids the standard UPSC close-option trap about Kidney nephron and homeostasis?	64
Q41. Which statement correctly identifies Nervous endocrine and feedback control?	65
Q42. Which option preserves the technical boundary of Nervous endocrine and feedback control?	65
Q43. Which statement uses Nervous endocrine and feedback control without changing its institution, unit or status?	66
Q44. Which option avoids the standard UPSC close-option trap about Nervous endocrine and feedback control?	66
Q45. Which statement correctly identifies Hormones metabolism and internal stability?	67
Q46. Which option preserves the technical boundary of Hormones metabolism and internal stability?	67
Q47. Which statement uses Hormones metabolism and internal stability without changing its institution, unit or status?	68
Q48. Which option avoids the standard UPSC close-option trap about Hormones metabolism and internal stability?	68
Q49. Which statement correctly identifies Innate adaptive and tolerance architecture?	69
Q50. Which option preserves the technical boundary of Innate adaptive and tolerance architecture?	69
Q51. Which statement uses Innate adaptive and tolerance architecture without changing its institution, unit or status?	70
Q52. Which option avoids the standard UPSC close-option trap about Innate adaptive and tolerance architecture?	70
Q53. Which statement correctly identifies Pathogen transmission vector and zoonosis map?	71
Q54. Which option preserves the technical boundary of Pathogen transmission vector and zoonosis map?	71
Q55. Which statement uses Pathogen transmission vector and zoonosis map without changing its institution, unit or status?	72
Q56. Which option avoids the standard UPSC close-option trap about Pathogen transmission vector and zoonosis map?	72
Q57. Which statement correctly identifies Antibiotic antiviral and AMR boundary?	73
Q58. Which option preserves the technical boundary of Antibiotic antiviral and AMR boundary?	73
Q59. Which statement uses Antibiotic antiviral and AMR boundary without changing its institution, unit or status?	74
Q60. Which option avoids the standard UPSC close-option trap about Antibiotic antiviral and AMR boundary?	74
Q61. Which statement correctly identifies Biofilms microbiome probiotics and extremophiles?	75
Q62. Which option preserves the technical boundary of Biofilms microbiome probiotics and extremophiles?	75
Q63. Which statement uses Biofilms microbiome probiotics and extremophiles without changing its institution, unit or status?	76
Q64. Which option avoids the standard UPSC close-option trap about Biofilms microbiome probiotics and extremophiles?	76
Q65. Which statement correctly identifies Plant physiology transport and stress response?	77
Q66. Which option preserves the technical boundary of Plant physiology transport and stress response?	77

Q67. Which statement uses Plant physiology transport and stress response without changing its institution, unit or status?	78
Q68. Which option avoids the standard UPSC close-option trap about Plant physiology transport and stress response?	78
Q69. Which statement correctly identifies Routed taxonomy and organism-form distinctions?	79
Q70. Which option preserves the technical boundary of Routed taxonomy and organism-form distinctions?	79
Q71. Which statement uses Routed taxonomy and organism-form distinctions without changing its institution, unit or status?	80
Q72. Which option avoids the standard UPSC close-option trap about Routed taxonomy and organism-form distinctions?	80
Q73. Which statement correctly identifies Cellulose decomposition route?	81
Q74. Which option preserves the technical boundary of Cellulose decomposition route?	81
Q75. Which statement uses Cellulose decomposition route without changing its institution, unit or status?	82
Q76. Which option avoids the standard UPSC close-option trap about Cellulose decomposition route?	82
Q77. Which statement correctly identifies Health institution and evidence-status firewall?	83
Q78. Which option preserves the technical boundary of Health institution and evidence-status firewall?	83
Q79. Which statement uses Health institution and evidence-status firewall without changing its institution, unit or status?	84
Q80. Which option avoids the standard UPSC close-option trap about Health institution and evidence-status firewall?	84
PYQS AND ANSWER PRACTICE	85
AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP	85
PYQ DEMAND CARD 1 - 2018-2023 Prelims GS-I	85
PYQ DEMAND CARD 2 - 2024-2025 Prelims GS-I	87
PYQ DEMAND CARD 3 - 2022 GS-III	88
ORIGINAL MAINS 1 - 10 MARKS	90
ORIGINAL MAINS 2 - 10 MARKS	93
ORIGINAL MAINS 3 - 15 MARKS	95
ORIGINAL MAINS 4 - 15 MARKS	99
ORIGINAL MAINS 5 - 20 MARKS	102
ORIGINAL MAINS 6 - 20 MARKS	105
OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER	108
CONSOLIDATED REGISTER NOTES	112
General Science: Biology and Physiology: CELL, BIOMOLECULE, GENETICS AND MEMBRANE RAPID MAP	112
General Science: Biology and Physiology: HUMAN PHYSIOLOGY, HOMEOSTASIS AND IMMUNITY SYSTEM GRID	114
General Science: Biology and Physiology: DISEASE, MICROBE, PLANT AND ROUTED-PYQ FIREWALLS	114
General Science: Biology and Physiology: PUBLIC-HEALTH INSTITUTION, STATUS AND ANSWER-WRITING SPINE	114
COMPLETE TOPIC ASCII MASTER FLOW DIAGRAM	114

General Science: Biology and Physiology - Learner-v2

Complete Learning Session

Authoring-only generation: 2026-09-04. No PDF was rendered and no tracker or index was mutated.

SOURCE, PROGRESSION AND CURRENT-LINKAGE AUDIT

- **Generation date:** 2026-09-04.
- **Repository-first evidence:** the Basic owner is taught first and preserved in full; the Advanced owner is retained only in the optional final teaching block.
- **OCR evidence:** Repository Markdown was primary. OCR-searchable official UPSC papers were supplementary only for printed routed demands. No answer key, performance value, procurement outcome, operational deployment, programme target or technology milestone was inferred.
- **Qdrant:** not used; repository Markdown and available OCR context were sufficient.
- **PYQ integrity:** Audited ledgers route thirteen historical Prelims demands from 2018-2023, six Prelims demands from 2024-2025 and the 2022 GS-III cellulose-decomposition demand to this owner. Three representative cards preserve historical objective, recent objective and Mains routes without reproducing answer keys. Vaccine-platform, monoclonal-antibody and gene-editing demands remain with Topics 15 and 14.
- **Live-link boundary:** Official-source checks on 2026-09-04 retained the 2024 microRNA citation, the 2025 peripheral-immune-tolerance citation, WHO's trachoma validation and polio-transition boundaries, and WHO/UNICEF's annual estimate methodology. No therapy approval, vaccine-platform claim, coverage percentage, eradication claim, medical advice or PYQ key was invented.
- **Fact/inference discipline:** no current-affairs item, PYQ wording, figure or quotation is invented.

LIVE OFFICIAL-SOURCE ATTEMPT LOG

The checks below were made on 2026-09-04. Substantive official text is used only for the proposition it supports; title-only, blocked or thin pages are recorded and supply no factual claim.

- <https://www.nobelprize.org/prizes/medicine/2024/press-release/> - fetched 2026-09-04; the official Nobel release confirmed the joint award to Victor Ambros and Gary Ruvkun for microRNA and its role in post-transcriptional gene regulation. The citation was used to anchor gene regulation, not to infer a diagnostic, therapy or policy outcome.
- <https://www.nobelprize.org/prizes/medicine/2025/press-release/> - fetched 2026-09-04; the official release confirmed the award to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance, including regulatory T cells and Foxp3. Clinical-trial references were not rewritten as approved treatments, population benefit or Indian programme outcomes.
- <https://www.who.int/news/item/08-10-2024-elimination-of-trachoma-as-a-public-health-problem-in-india> - fetched 2026-09-04; WHO confirmed validation of India's elimination of trachoma as a public-health problem, the bacterial cause, transmission routes and SAFE strategy. Validation was not presented as eradication of the organism or as an end to post-validation surveillance.
- <https://www.who.int/publications/m/item/india-polio-transition-snapshot> - fetched 2026-09-04; WHO confirmed that India's transition approach seeks to sustain polio-free status and reuse the eradication network for immunization, surveillance and emergency response. The page supplied no basis for a new coverage or disease-incidence claim.
- <https://www.who.int/teams/immunization-vaccines-and-biologicals/immunization-analysis-and-insights/global-monitoring/immunization-coverage/who-unicef-estimates-of-national-immunization-coverage> - fetched 2026-09-04; WHO confirmed that estimates use Member-State and other reviewed data, are updated annually, and then reflected data reported through 21 June 2026. No country percentage was imported.

BASIC LEARNING SESSION

DEEP-REVIEW LEARNING CONTRACT

Control	Binding rule for this package
Syllabus boundary	Complete Science and Technology Basic/Core is easy-first and answer-complete before optional Advanced depth.
System boundary	Organisation, platform, subsystem, payload, process, application, institution and outcome remain distinct.
Technical boundary	Physical mechanism, classification, stage, platform, unit, range/capacity and limiting condition are explicit.
Status boundary	Announcement, approval, development, test, deployment, induction, commercial operation and observed outcome remain distinct.
Data boundary	Every mission, launch, target, capacity, budget, range, timeline, rule and regulatory claim keeps source, date, unit and status.
Governance boundary	Funder, developer, operator, authoriser, regulator, procurer, settlement actor and adjudicator are not conflated.
Practice contract	Every solved item has demand decoding, detailed examiner-grade model, executable timed/compression plan, marks rationale and answer-specific improvement.
PYQ contract	Official wording and keys are asserted only when verified; routed or reconstructed demands remain labelled.
Dual-flow contract	Graphical and ASCII masters independently reconstruct the same complete Basic-to-optional-Advanced evidence spine.
Approval	This immutable successor remains approved: false pending explicit approval.

Canonical Basic/Core owner: upsc-ai-kit\knowledge\Science-and-Technology\basic\23_General-Science-Biology-and-Physiology.md **Substantive canonical provenance owner:** upsc-ai-kit\knowledge\Science-and-Technology\learning-sessions\v2\subject-wide-syllabus\science-and-technology-23_Learning-Session.md **Optional Advanced owner:** upsc-ai-kit\knowledge\Science-and-Technology\advanced\23_General-Science-Biology-and-Physiology.md **Official syllabus mapping:** upsc-ai-kit\knowledge\Science-and-Technology\OFFICIAL-UPSC-SYLLABUS-MAPPING.md

EVIDENCE, PYQ AND CURRENT-STATUS CONTROL

- ISRO organisation, launcher stages, payload and orbit claims retain their exact object and mission date.
- NavIC, GAGAN, human-spaceflight, planetary, nuclear, fusion, missile and procurement claims retain category and status.
- Aadhaar/UPI, AI, quantum and semiconductor claims retain actor, layer, metric, maturity and governance boundaries.
- DPDP/cybersecurity, biotechnology and genetic-engineering claims retain legal/regulatory stage and competent institution.
- Targets, budgets, capacities, ranges and timelines are never promoted into achieved outcomes without dated evidence.
- **Current-status note, rechecked 2026-09-06:** volatile claims retain source/date/status; failed official fetches remain explicit and supply no invented fact.

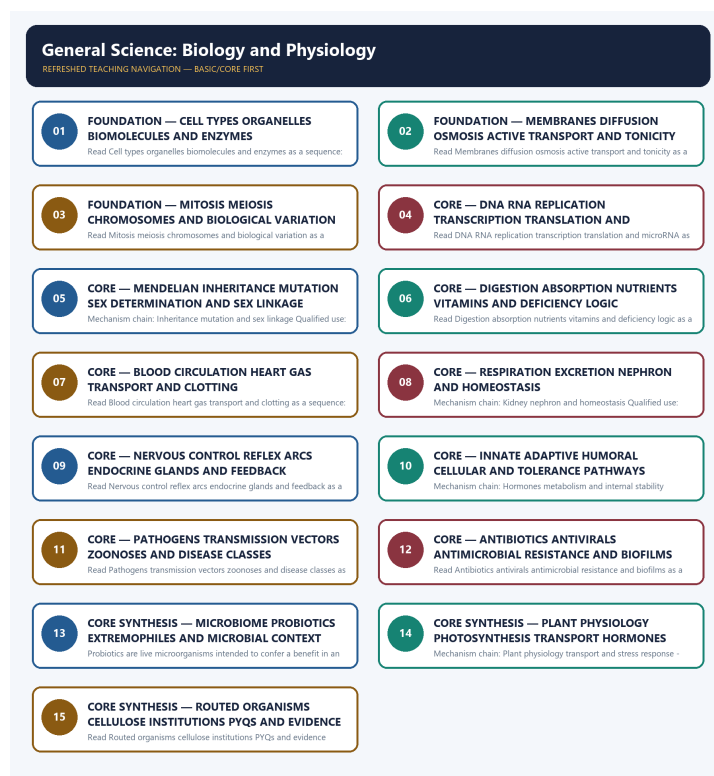
Generation-local live/current sources:

- <https://www.nobelprize.org/prizes/medicine/2024/press-release/> - fetched 2026-09-04; the official Nobel release confirmed the joint award to Victor Ambros and Gary Ruvkun for microRNA and its role in post-transcriptional gene regulation. The citation was used to anchor gene regulation, not to infer a diagnostic, therapy or policy outcome.

- <https://www.nobelprize.org/prizes/medicine/2025/press-release/> - fetched 2026-09-04; the official release confirmed the award to Mary E. Brunkow, Fred Ramsdell and Shimon Sakaguchi for discoveries concerning peripheral immune tolerance, including regulatory T cells and Foxp3. Clinical-trial references were not rewritten as approved treatments, population benefit or Indian programme outcomes.
- <https://www.who.int/news/item/08-10-2024-elimination-of-trachoma-as-a-public-health-problem-in-india> - fetched 2026-09-04; WHO confirmed validation of India's elimination of trachoma as a public-health problem, the bacterial cause, transmission routes and SAFE strategy. Validation was not presented as eradication of the organism or as an end to post-validation surveillance.
- <https://www.who.int/publications/m/item/india-polio-transition-snapshot> - fetched 2026-09-04; WHO confirmed that India's transition approach seeks to sustain polio-free status and reuse the eradication network for immunization, surveillance and emergency response. The page supplied no basis for a new coverage or disease-incidence claim.
- <https://www.who.int/teams/immunization-vaccines-and-biologicals/immunization-analysis-and-insights/global-monitoring/immunization-coverage/who-unicef-estimates-of-national-immunization-coverage> - fetched 2026-09-04; WHO confirmed that estimates use Member-State and other reviewed data, are updated annually, and then reflected data reported through 21 June 2026. No country percentage was imported.

Topic-routed verified PYQ owners:

- No topic-local official question file is claimed; repository PYQ routing ledgers control wording and status.



Refreshed teaching navigation

Distinct embedded teaching-navigation image. The separate continuous Cārvāka-style flowchart package remains an independent at-a-glance artifact.

SESSION 1 - FOUNDATION - CELL TYPES ORGANELLES BIOMOLECULES AND ENZYMES

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them.

Technical definition: Read Cell types organelles biomolecules and enzymes as a sequence: identify Cell organisation and organelle boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Cell types organelles biomolecules and enzymes as a sequence: identify Cell organisation and organelle boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- FOUNDATION
- Cell types organelles biomolecules
- enzymes
- Mechanism chain
- Qualified use
- A cell

How to use them: Frame the answer through FOUNDATION; define Cell types organelles biomolecules, connect enzymes with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

CELL TYPES ORGANELLES BIOMOLECULES AND ENZYMES

01. Cell organisation and organelle boundary

|
v

02. Biomolecules and enzyme specificity

STATUS / CATEGORY FIREWALL -> Do not merge prokaryotic and eukaryotic cells or assign every organelle to every cell.

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function.

SIMPLE SYSTEM EXAMPLE

Read Cell types organelles biomolecules and enzymes as a sequence: identify Cell organisation and organelle boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not merge prokaryotic and eukaryotic cells or assign every organelle to every cell. This keeps Cell organisation and organelle boundary and Biomolecules and enzyme specificity in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole.
- Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function.

EXAMINER CAUTION

- Do not merge prokaryotic and eukaryotic cells or assign every organelle to every cell.

EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Move from cell type to organelle, biomolecule, function and condition-specific limitation.

MINI RECAP

- **Mechanism chain:** Cell organisation and organelle boundary -> Biomolecules and enzyme specificity
- **Qualified use:** Move from cell type to organelle, biomolecule, function and condition-specific limitation.

CLOSING RECALL FLOW - FOUNDATION - CELL TYPES ORGANELLES BIOMOLECULES AND ENZYMES

START / CONCEPT: FOUNDATION - Cell types organelles biomolecules and enzymes

|

v

EXACT TERMS: FOUNDATION · Cell types organelles biomolecules · enzymes · Mechanism chain ·
Qualified use · A cell

|

v

MECHANISM / ARGUMENT: Mechanism chain: Cell organisation and organelle boundary - Biomolecules and enzyme specificity Qualified use: Move from cell type to organelle, biomolecule, function and condition-specific limitation.

|

v

CONSEQUENCE / CONTRAST: A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them.

|

v

UPSC TRAP / ANSWER-USE: Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function.

|

v

ANSWER-GRABBING FORMULATION: Read Cell types organelles biomolecules and enzymes as a sequence: identify Cell organisation and organelle boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

SESSION 2 - FOUNDATION - MEMBRANES DIFFUSION OSMOSIS ACTIVE TRANSPORT AND TONICITY

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient.

Technical definition: Read Membranes diffusion osmosis active transport and tonicity as a sequence: identify Membrane transport and water balance, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient.

MUST-WRITE KEYWORDS

- FOUNDATION
- Membranes diffusion osmosis active transport
- tonicity
- Mechanism chain
- Qualified use
- Diffusion

How to use them: Frame the answer through FOUNDATION; define Membranes diffusion osmosis active transport, connect tonicity with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

MEMBRANES DIFFUSION OSMOSIS ACTIVE TRANSPORT AND TONICITY

01. Membrane transport and water balance

STATUS / CATEGORY FIREWALL -> Do not give every carbohydrate, lipid, protein or enzyme one universal function.

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context.

SIMPLE SYSTEM EXAMPLE

Read Membranes diffusion osmosis active transport and tonicity as a sequence: identify Membrane transport and water balance, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not give every carbohydrate, lipid, protein or enzyme one universal function. This keeps Membrane transport and water balance in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context.

EXAMINER CAUTION

- Do not give every carbohydrate, lipid, protein or enzyme one universal function.

EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Fix the membrane, gradient, water or solute and energy requirement before predicting movement.

MINI RECAP

- **Mechanism chain:** Membrane transport and water balance
- **Qualified use:** Fix the membrane, gradient, water or solute and energy requirement before predicting movement.

CLOSING RECALL FLOW - FOUNDATION - MEMBRANES DIFFUSION OSMOSIS ACTIVE TRANSPORT AND TONICITY

START / CONCEPT: FOUNDATION - Membranes diffusion osmosis active transport and tonicity

|
v

EXACT TERMS: FOUNDATION · Membranes diffusion osmosis active transport · tonicity · Mechanism chain
· Qualified use · Diffusion

|
v

MECHANISM / ARGUMENT: Read Membranes diffusion osmosis active transport and tonicity as a sequence: identify Membrane transport and water balance, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

|
v

CONSEQUENCE / CONTRAST: Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context.

|
v

UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: membrane transport and water balance Qualified use.

|
v

ANSWER-GRABBING FORMULATION: Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient.

SESSION 3 - FOUNDATION - MITOSIS MEIOSIS CHROMOSOMES AND BIOLOGICAL VARIATION

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation.

Technical definition: Read Mitosis meiosis chromosomes and biological variation as a sequence: identify Mitosis meiosis and variation, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Mitosis meiosis chromosomes and biological variation as a sequence: identify Mitosis meiosis and variation, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **FOUNDATION**
- **Mitosis meiosis chromosomes**
- **biological variation**

- **Mechanism chain**
- **Qualified use**
- **Mitosis**

How to use them: Frame the answer through FOUNDATION; define Mitosis meiosis chromosomes, connect biological variation with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

MITOSIS MEIOSIS CHROMOSOMES AND BIOLOGICAL VARIATION

01. Mitosis meiosis and variation

STATUS / CATEGORY FIREWALL -> Do not confuse diffusion, osmosis and energy-dependent active transport.

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell.

SIMPLE SYSTEM EXAMPLE

Read Mitosis meiosis chromosomes and biological variation as a sequence: identify Mitosis meiosis and variation, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not confuse diffusion, osmosis and energy-dependent active transport. This keeps Mitosis meiosis and variation in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell.

EXAMINER CAUTION

- Do not confuse diffusion, osmosis and energy-dependent active transport.

EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Track division count, chromosome state, product number and source of variation.

MINI RECAP

- **Mechanism chain:** Mitosis meiosis and variation
- **Qualified use:** Track division count, chromosome state, product number and source of variation.

CLOSING RECALL FLOW - FOUNDATION - MITOSIS MEIOSIS CHROMOSOMES AND BIOLOGICAL VARIATION

START / CONCEPT: FOUNDATION - Mitosis meiosis chromosomes and biological variation

|
v

EXACT TERMS: FOUNDATION · Mitosis meiosis chromosomes · biological variation · Mechanism chain · Qualified use · Mitosis

|
v

MECHANISM / ARGUMENT: Mechanism chain: Mitosis meiosis and variation Qualified use: Track division count, chromosome state, product number and source of variation.

|
v

CONSEQUENCE / CONTRAST: Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation.

|
v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not confuse diffusion, osmosis and energy-dependent active transport.

|
v

ANSWER-GRABBING FORMULATION: Read Mitosis meiosis chromosomes and biological variation as a sequence: identify Mitosis meiosis and variation, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

SESSION 4 - CORE - DNA RNA REPLICATION TRANSCRIPTION TRANSLATION AND MICRORNA

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages.

Technical definition: Read DNA RNA replication transcription translation and microRNA as a sequence: identify DNA RNA gene expression and microRNA, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Mechanism chain: DNA RNA gene expression and microRNA Qualified use: Trace information from DNA through RNA to protein while preserving regulatory RNA roles.

MUST-WRITE KEYWORDS

- DNA RNA replication transcription translation
- microRNA
- Mechanism chain
- Qualified use
- DNA
- RNA

How to use them: Frame the answer through DNA RNA replication transcription translation; define microRNA, connect Mechanism chain with Qualified use to explain the mechanism, and use DNA for the decisive comparison or qualification.

VISUAL FIRST

DNA RNA REPLICATION TRANSCRIPTION TRANSLATION AND MICRORNA

01. DNA RNA gene expression and microRNA

STATUS / CATEGORY FIREWALL -> Do not merge mitosis with meiosis or crossing over with gene editing.

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages.

SIMPLE SYSTEM EXAMPLE

Read DNA RNA replication transcription translation and microRNA as a sequence: identify DNA RNA gene expression and microRNA, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not merge mitosis with meiosis or crossing over with gene editing. This keeps DNA RNA gene expression and microRNA in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages.

EXAMINER CAUTION

- Do not merge mitosis with meiosis or crossing over with gene editing.

EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Trace information from DNA through RNA to protein while preserving regulatory RNA roles.

MINI RECAP

- **Mechanism chain:** DNA RNA gene expression and microRNA
- **Qualified use:** Trace information from DNA through RNA to protein while preserving regulatory RNA roles.

CLOSING RECALL FLOW - CORE - DNA RNA REPLICATION TRANSCRIPTION TRANSLATION AND MICRORNA

START / CONCEPT: CORE - DNA RNA replication transcription translation and microRNA

|
v

EXACT TERMS: DNA RNA replication transcription translation · microRNA · Mechanism chain · Qualified use · DNA · RNA

|
v

MECHANISM / ARGUMENT: Read DNA RNA replication transcription translation and microRNA as a sequence: identify DNA RNA gene expression and microRNA, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

|
v

CONSEQUENCE / CONTRAST: DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons.

|
v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not merge mitosis with meiosis or crossing over with gene editing.

|
v

ANSWER-GRABBING FORMULATION: Mechanism chain: DNA RNA gene expression and microRNA Qualified use: Trace information from DNA through RNA to protein while preserving regulatory RNA roles.

SESSION 5 - CORE - MENDELIAN INHERITANCE MUTATION SEX DETERMINATION AND SEX LINKAGE

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Read Mendelian inheritance mutation sex determination and sex linkage as a sequence: identify Inheritance mutation and sex linkage, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

Technical definition: Mechanism chain: Inheritance mutation and sex linkage Qualified use: Define gene and allele, apply segregation carefully and qualify mutation or sex-linked inference.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns.

MUST-WRITE KEYWORDS

- Mendelian inheritance mutation sex determination
- sex linkage
- Mechanism chain
- Qualified use
- A gene
- DNA

How to use them: Frame the answer through Mendelian inheritance mutation sex determination; define sex linkage, connect Mechanism chain with Qualified use to explain the mechanism, and use A gene for the decisive comparison or qualification.

VISUAL FIRST

MENDELIAN INHERITANCE MUTATION SEX DETERMINATION AND SEX LINKAGE

01. Inheritance mutation and sex linkage

STATUS / CATEGORY FIREWALL -> Do not reduce RNA to a passive messenger or equate transcription with translation.

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns.

SIMPLE SYSTEM EXAMPLE

Read Mendelian inheritance mutation sex determination and sex linkage as a sequence: identify Inheritance mutation and sex linkage, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not reduce RNA to a passive messenger or equate transcription with translation. This keeps Inheritance mutation and sex linkage in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns.

EXAMINER CAUTION

- Do not reduce RNA to a passive messenger or equate transcription with translation.

EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Define gene and allele, apply segregation carefully and qualify mutation or sex-linked inference.

MINI RECAP

- **Mechanism chain:** Inheritance mutation and sex linkage
- **Qualified use:** Define gene and allele, apply segregation carefully and qualify mutation or sex-linked inference.

CLOSING RECALL FLOW - CORE - MENDELIAN INHERITANCE MUTATION SEX DETERMINATION AND SEX LINKAGE

START / CONCEPT: CORE - Mendelian inheritance mutation sex determination and sex linkage
|
v
EXACT TERMS: Mendelian inheritance mutation sex determination · sex linkage · Mechanism chain ·
Qualified use · A gene · DNA
|
v
MECHANISM / ARGUMENT: Read Mendelian inheritance mutation sex determination and sex linkage as a
sequence: identify Inheritance mutation and sex linkage, follow the named mechanism to its user or
output, and stop at the last verified status rather than assuming the next stage.
|
v
CONSEQUENCE / CONTRAST: Mechanism chain: Inheritance mutation and sex linkage Qualified use: Define
gene and allele, apply segregation carefully and qualify mutation or sex-linked inference.
|
v
UPSC TRAP / ANSWER-USE: The decisive boundary is Do not reduce RNA to a passive messenger or equate
transcription with translation.
|
v
ANSWER-GRABBING FORMULATION: Mutations may be point, frameshift or chromosomal changes and supply
variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive
inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns.

SESSION 6 - CORE - DIGESTION ABSORPTION NUTRIENTS VITAMINS AND DEFICIENCY LOGIC

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.

Technical definition: Read Digestion absorption nutrients vitamins and deficiency logic as a sequence: identify Digestion absorption and nutrition, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Digestion absorption nutrients vitamins and deficiency logic as a sequence: identify Digestion absorption and nutrition, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- Digestion absorption nutrients vitamins
- deficiency logic
- Mechanism chain
- Qualified use
- Digestion
- Vitamins

How to use them: Frame the answer through Digestion absorption nutrients vitamins; define deficiency logic, connect Mechanism chain with Qualified use to explain the mechanism, and use Digestion for the decisive comparison or qualification.

VISUAL FIRST

DIGESTION ABSORPTION NUTRIENTS VITAMINS AND DEFICIENCY LOGIC

01. Digestion absorption and nutrition

|

v

02. Blood heart and double circulation

STATUS / CATEGORY FIREWALL -> Do not treat a mutation, gene association or sex-linked pattern as deterministic destiny.

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions.

SIMPLE SYSTEM EXAMPLE

Read Digestion absorption nutrients vitamins and deficiency logic as a sequence: identify Digestion absorption and nutrition, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not treat a mutation, gene association or sex-linked pattern as deterministic destiny. This keeps Digestion absorption and nutrition and Blood heart and double circulation in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.
- Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions.

EXAMINER CAUTION

- Do not treat a mutation, gene association or sex-linked pattern as deterministic destiny.

EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Follow food breakdown to enzyme condition, absorption site, nutrient function and deficiency mechanism.

MINI RECAP

- **Mechanism chain:** Digestion absorption and nutrition -> Blood heart and double circulation
- **Qualified use:** Follow food breakdown to enzyme condition, absorption site, nutrient function and deficiency mechanism.

CLOSING RECALL FLOW - CORE - DIGESTION ABSORPTION NUTRIENTS VITAMINS AND DEFICIENCY LOGIC

START / CONCEPT: CORE - Digestion absorption nutrients vitamins and deficiency logic

|
v

EXACT TERMS: Digestion absorption nutrients vitamins · deficiency logic · Mechanism chain ·
Qualified use · Digestion · Vitamins

|
v

MECHANISM / ARGUMENT: Mechanism chain: Digestion absorption and nutrition - Blood heart and double
circulation Qualified use: Follow food breakdown to enzyme condition, absorption site, nutrient
function and deficiency mechanism.

|
v

CONSEQUENCE / CONTRAST: Vitamins, minerals, protein and energy have different roles; deficiency,
absorption failure and excess are separate nutritional problems.

|
v

UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: follow food breakdown to enzyme
condition, absorption site, nutrient function and deficiency mechanism.

|
v

ANSWER-GRABBING FORMULATION: Read Digestion absorption nutrients vitamins and deficiency logic as a
sequence: identify Digestion absorption and nutrition, follow the named mechanism to its user or
output, and stop at the last verified status rather than assuming the next stage.

SESSION 7 - CORE - BLOOD CIRCULATION HEART GAS TRANSPORT AND CLOTTING

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous.

Technical definition: Read Blood circulation heart gas transport and clotting as a sequence: identify Respiration gas exchange and cellular energy, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Blood circulation heart gas transport and clotting as a sequence: identify Respiration gas exchange and cellular energy, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- Blood circulation heart gas transport
- clotting
- Mechanism chain
- Qualified use
- Ventilation
- Haemoglobin

How to use them: Frame the answer through Blood circulation heart gas transport; define clotting, connect Mechanism chain with Qualified use to explain the mechanism, and use Ventilation for the decisive comparison or qualification.

VISUAL FIRST

BLOOD CIRCULATION HEART GAS TRANSPORT AND CLOTTING

01. Respiration gas exchange and cellular energy

STATUS / CATEGORY FIREWALL -> Do not call bile an enzyme or the stomach the principal site of nutrient absorption.

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous.

SIMPLE SYSTEM EXAMPLE

Read Blood circulation heart gas transport and clotting as a sequence: identify Respiration gas exchange and cellular energy, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not call bile an enzyme or the stomach the principal site of nutrient absorption. This keeps Respiration gas exchange and cellular energy in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous.

EXAMINER CAUTION

- Do not call bile an enzyme or the stomach the principal site of nutrient absorption.

EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Separate blood component, vessel direction, heart circuit, gas carriage and clotting role.

MINI RECAP

- **Mechanism chain:** Respiration gas exchange and cellular energy
- **Qualified use:** Separate blood component, vessel direction, heart circuit, gas carriage and clotting role.

CLOSING RECALL FLOW - CORE - BLOOD CIRCULATION HEART GAS TRANSPORT AND CLOTTING

START / CONCEPT: CORE - Blood circulation heart gas transport and clotting

|

v

EXACT TERMS: Blood circulation heart gas transport · clotting · Mechanism chain · Qualified use · Ventilation · Haemoglobin

|

v

MECHANISM / ARGUMENT: Mechanism chain: Respiration gas exchange and cellular energy Qualified use: Separate blood component, vessel direction, heart circuit, gas carriage and clotting role.

|

v

CONSEQUENCE / CONTRAST: Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous.

|

v

UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces...

|

v

ANSWER-GRABBING FORMULATION: Read Blood circulation heart gas transport and clotting as a sequence: identify Respiration gas exchange and cellular energy, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

SESSION 8 - CORE - RESPIRATION EXCRETION NEPHRON AND HOMEOSTASIS

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Read Respiration excretion nephron and homeostasis as a sequence: identify Kidney nephron and homeostasis, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

Technical definition: Mechanism chain: Kidney nephron and homeostasis Qualified use: Distinguish ventilation, gas exchange, cellular energy, filtration and whole-body regulation.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Respiration excretion nephron and homeostasis as a sequence: identify Kidney nephron and homeostasis, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- Respiration excretion nephron
- homeostasis
- Mechanism chain
- Qualified use
- Dialysis
- Read Respiration

How to use them: Frame the answer through Respiration excretion nephron; define homeostasis, connect Mechanism chain with Qualified use to explain the mechanism, and use Dialysis for the decisive comparison or qualification.

VISUAL FIRST

RESPIRATION EXCRETION NEPHRON AND HOMEOSTASIS

01. Kidney nephron and homeostasis

STATUS / CATEGORY FIREWALL -> Do not define arteries and veins by oxygen content.

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys.

SIMPLE SYSTEM EXAMPLE

Read Respiration excretion nephron and homeostasis as a sequence: identify Kidney nephron and homeostasis, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not define arteries and veins by oxygen content. This keeps Kidney nephron and homeostasis in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys.

EXAMINER CAUTION

- Do not define arteries and veins by oxygen content.

EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Distinguish ventilation, gas exchange, cellular energy, filtration and whole-body regulation.

MINI RECAP

- **Mechanism chain:** Kidney nephron and homeostasis
- **Qualified use:** Distinguish ventilation, gas exchange, cellular energy, filtration and whole-body regulation.

CLOSING RECALL FLOW - CORE - RESPIRATION EXCRETION NEPHRON AND HOMEOSTASIS

START / CONCEPT: CORE - Respiration excretion nephron and homeostasis

|

v

EXACT TERMS: Respiration excretion nephron · homeostasis · Mechanism chain · Qualified use ·

Dialysis · Read Respiration

|

v

MECHANISM / ARGUMENT: Mechanism chain: Kidney nephron and homeostasis Qualified use: Distinguish ventilation, gas exchange, cellular energy, filtration and whole-body regulation.

|

v

CONSEQUENCE / CONTRAST: Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys.

|

v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not define arteries and veins by oxygen content.

|

v

ANSWER-GRABBING FORMULATION: Read Respiration excretion nephron and homeostasis as a sequence: identify Kidney nephron and homeostasis, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

SESSION 9 - CORE - NERVOUS CONTROL REFLEX ARCS ENDOCRINE GLANDS AND FEEDBACK

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems.

Technical definition: Read Nervous control reflex arcs endocrine glands and feedback as a sequence: identify Nervous endocrine and feedback control, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Nervous control reflex arcs endocrine glands and feedback as a sequence: identify Nervous endocrine and feedback control, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- Nervous control reflex arcs endocrine glands
- feedback
- Mechanism chain
- Qualified use
- Endocrine
- Read Nervous

How to use them: Frame the answer through Nervous control reflex arcs endocrine glands; define feedback, connect Mechanism chain with Qualified use to explain the mechanism, and use Endocrine for the decisive comparison or qualification.

VISUAL FIRST

NERVOUS CONTROL REFLEX ARCS ENDOCRINE GLANDS AND FEEDBACK

01. Nervous endocrine and feedback control

STATUS / CATEGORY FIREWALL -> Do not use breathing, alveolar gas exchange and cellular respiration as synonyms.

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems.

SIMPLE SYSTEM EXAMPLE

Read Nervous control reflex arcs endocrine glands and feedback as a sequence: identify Nervous endocrine and feedback control, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not use breathing, alveolar gas exchange and cellular respiration as synonyms. This keeps Nervous endocrine and feedback control in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems.

EXAMINER CAUTION

- Do not use breathing, alveolar gas exchange and cellular respiration as synonyms.

EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Map receptor, signal route, effector and negative feedback across neural and hormonal control.

MINI RECAP

- **Mechanism chain:** Nervous endocrine and feedback control
- **Qualified use:** Map receptor, signal route, effector and negative feedback across neural and hormonal control.

CLOSING RECALL FLOW - CORE - NERVOUS CONTROL REFLEX ARCS ENDOCRINE GLANDS AND FEEDBACK

START / CONCEPT: CORE - Nervous control reflex arcs endocrine glands and feedback

|

v

EXACT TERMS: Nervous control reflex arcs endocrine glands · feedback · Mechanism chain · Qualified use · Endocrine · Read Nervous

|

v

MECHANISM / ARGUMENT: Mechanism chain: Nervous endocrine and feedback control Qualified use: Map receptor, signal route, effector and negative feedback across neural and hormonal control.

|

v

CONSEQUENCE / CONTRAST: Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems.

|

v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not use breathing, alveolar gas exchange and cellular respiration as synonyms.

|

v

ANSWER-GRABBING FORMULATION: Read Nervous control reflex arcs endocrine glands and feedback as a sequence: identify Nervous endocrine and feedback control, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

SESSION 10 - CORE - INNATE ADAPTIVE HUMORAL CELLULAR AND TOLERANCE PATHWAYS

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Read Innate adaptive humoral cellular and tolerance pathways as a sequence: identify Hormones metabolism and internal stability, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

Technical definition: Mechanism chain: Hormones metabolism and internal stability Qualified use: Classify barrier, cell, antibody, memory and tolerance before making a public-health claim.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Innate adaptive humoral cellular and tolerance pathways as a sequence: identify Hormones metabolism and internal stability, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- Innate adaptive humoral cellular
- tolerance pathways
- Mechanism chain
- Qualified use
- Homeostasis
- Read Innate

How to use them: Frame the answer through Innate adaptive humoral cellular; define tolerance pathways, connect Mechanism chain with Qualified use to explain the mechanism, and use Homeostasis for the decisive comparison or qualification.

VISUAL FIRST

INNATE ADAPTIVE HUMORAL CELLULAR AND TOLERANCE PATHWAYS

01. Hormones metabolism and internal stability

STATUS / CATEGORY FIREWALL -> Do not claim dialysis reproduces every kidney function.

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms.

SIMPLE SYSTEM EXAMPLE

Read Innate adaptive humoral cellular and tolerance pathways as a sequence: identify Hormones metabolism and internal stability, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not claim dialysis reproduces every kidney function. This keeps Hormones metabolism and internal stability in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms.

EXAMINER CAUTION

- Do not claim dialysis reproduces every kidney function.

EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Classify barrier, cell, antibody, memory and tolerance before making a public-health claim.

MINI RECAP

- **Mechanism chain:** Hormones metabolism and internal stability
- **Qualified use:** Classify barrier, cell, antibody, memory and tolerance before making a public-health claim.

CLOSING RECALL FLOW - CORE - INNATE ADAPTIVE HUMORAL CELLULAR AND TOLERANCE PATHWAYS

START / CONCEPT: CORE - Innate adaptive humoral cellular and tolerance pathways

|
v

EXACT TERMS: Innate adaptive humoral cellular · tolerance pathways · Mechanism chain · Qualified use · Homeostasis · Read Innate

|
v

MECHANISM / ARGUMENT: Mechanism chain: Hormones metabolism and internal stability Qualified use: Classify barrier, cell, antibody, memory and tolerance before making a public-health claim.

|
v

CONSEQUENCE / CONTRAST: Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms.

|
v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not claim dialysis reproduces every kidney function.

|
v

ANSWER-GRABBING FORMULATION: Read Innate adaptive humoral cellular and tolerance pathways as a sequence: identify Hormones metabolism and internal stability, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

SESSION 11 - CORE - PATHOGENS TRANSMISSION VECTORS ZONOOSES AND DISEASE CLASSES

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself.

Technical definition: Read Pathogens transmission vectors zoonoses and disease classes as a sequence: identify Innate adaptive and tolerance architecture, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself.

MUST-WRITE KEYWORDS

- Pathogens transmission vectors zoonoses
- disease classes
- Mechanism chain
- Qualified use
- Innate
- B-cell

How to use them: Frame the answer through Pathogens transmission vectors zoonoses; define disease classes, connect Mechanism chain with Qualified use to explain the mechanism, and use Innate for the decisive comparison or qualification.

VISUAL FIRST

PATHOGENS TRANSMISSION VECTORS ZOONOSIS AND DISEASE CLASSES

01. Innate adaptive and tolerance architecture

|
v

02. Pathogen transmission vector and zoonosis map

STATUS / CATEGORY FIREWALL -> Do not merge a neural impulse, neurotransmitter, enzyme and hormone.

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself.

SIMPLE SYSTEM EXAMPLE

Read Pathogens transmission vectors zoonoses and disease classes as a sequence: identify Innate adaptive and tolerance architecture, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not merge a neural impulse, neurotransmitter, enzyme and hormone. This keeps Innate adaptive and tolerance architecture and Pathogen transmission vector and zoonosis map in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone.
- Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself.

EXAMINER CAUTION

- Do not merge a neural impulse, neurotransmitter, enzyme and hormone.

EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Name the agent, reservoir, route, vector and prevention layer without category drift.

MINI RECAP

- **Mechanism chain:** Innate adaptive and tolerance architecture -> Pathogen transmission vector and zoonosis map
- **Qualified use:** Name the agent, reservoir, route, vector and prevention layer without category drift.

CLOSING RECALL FLOW - CORE - PATHOGENS TRANSMISSION VECTORS ZOOSES AND DISEASE CLASSES

START / CONCEPT: CORE - Pathogens transmission vectors zoonoses and disease classes

|
v

EXACT TERMS: Pathogens transmission vectors zoonoses · disease classes · Mechanism chain · Qualified use · Innate · B-cell

|
v

MECHANISM / ARGUMENT: Read Pathogens transmission vectors zoonoses and disease classes as a sequence: identify Innate adaptive and tolerance architecture, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

|
v

CONSEQUENCE / CONTRAST: Mechanism chain: Innate adaptive and tolerance architecture - Pathogen transmission vector and zoonosis map Qualified use: Name the agent, reservoir, route, vector and prevention layer without category drift.

|
v

UPSC TRAP / ANSWER-USE: B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone.

|
v

ANSWER-GRABBING FORMULATION: Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself.

SESSION 12 - CORE - ANTIBIOTICS ANTIVIRALS ANTIMICROBIAL RESISTANCE AND BIOFILMS

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body.

Technical definition: Read Antibiotics antivirals antimicrobial resistance and biofilms as a sequence: identify Antibiotic antiviral and AMR boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Antibiotics antivirals antimicrobial resistance and biofilms as a sequence: identify Antibiotic antiviral and AMR boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- Antibiotics antivirals antimicrobial resistance
- biofilms
- Mechanism chain
- Qualified use
- Antimicrobial resistance
- Antimicrobial

How to use them: Frame the answer through Antibiotics antivirals antimicrobial resistance; define biofilms, connect Mechanism chain with Qualified use to explain the mechanism, and use Antimicrobial resistance for the decisive comparison or qualification.

VISUAL FIRST

ANTIBIOTICS ANTIVIRALS ANTIMICROBIAL RESISTANCE AND BIOFILMS

01. Antibiotic antiviral and AMR boundary

STATUS / CATEGORY FIREWALL -> Do not describe homeostasis as a perfectly fixed internal value.

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body.

SIMPLE SYSTEM EXAMPLE

Read Antibiotics antivirals antimicrobial resistance and biofilms as a sequence: identify Antibiotic antiviral and AMR boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not describe homeostasis as a perfectly fixed internal value. This keeps Antibiotic antiviral and AMR boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body.

EXAMINER CAUTION

- Do not describe homeostasis as a perfectly fixed internal value.

EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Match the drug class to the organism and explain resistance as microbial selection.

MINI RECAP

- **Mechanism chain:** Antibiotic antiviral and AMR boundary
- **Qualified use:** Match the drug class to the organism and explain resistance as microbial selection.

CLOSING RECALL FLOW - CORE - ANTIBIOTICS ANTIVIRALS ANTIMICROBIAL RESISTANCE AND BIOFILMS

START / CONCEPT: CORE - Antibiotics antivirals antimicrobial resistance and biofilms

|
v

EXACT TERMS: Antibiotics antivirals antimicrobial resistance · biofilms · Mechanism chain · Qualified use · Antimicrobial resistance · Antimicrobial

|
v

MECHANISM / ARGUMENT: Mechanism chain: Antibiotic antiviral and AMR boundary Qualified use: Match the drug class to the organism and explain resistance as microbial selection.

|
v

CONSEQUENCE / CONTRAST: Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body.

|
v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not describe homeostasis as a perfectly fixed internal value.

|
v

ANSWER-GRABBING FORMULATION: Read Antibiotics antivirals antimicrobial resistance and biofilms as a sequence: identify Antibiotic antiviral and AMR boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

SESSION 13 - CORE SYNTHESIS - MICROBIOME PROBIOTICS EXTREMOPHILES AND MICROBIAL CONTEXT

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion.

Technical definition: Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion.

MUST-WRITE KEYWORDS

- CORE SYNTHESIS
- Microbiome probiotics extremophiles
- microbial context
- Mechanism chain
- Qualified use
- A biofilm

How to use them: Frame the answer through CORE SYNTHESIS; define Microbiome probiotics extremophiles, connect microbial context with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

MICROBIOME PROBIOTICS EXTREMOPHILES AND MICROBIAL CONTEXT

01. Biofilms microbiome probiotics and extremophiles

STATUS / CATEGORY FIREWALL -> Do not equate immunity with antibodies alone or an antigen with an antibody.

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context.

SIMPLE SYSTEM EXAMPLE

Read Microbiome probiotics extremophiles and microbial context as a sequence: identify Biofilms microbiome probiotics and extremophiles, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not equate immunity with antibodies alone or an antigen with an antibody. This keeps Biofilms microbiome probiotics and extremophiles in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context.

EXAMINER CAUTION

- Do not equate immunity with antibodies alone or an antigen with an antibody.

EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** State community, host or environmental context before inferring benefit, hazard or treatment response.

MINI RECAP

- **Mechanism chain:** Biofilms microbiome probiotics and extremophiles
- **Qualified use:** State community, host or environmental context before inferring benefit, hazard or treatment response.

CLOSING RECALL FLOW - CORE SYNTHESIS - MICROBIOME PROBIOTICS EXTREMOPHILES AND MICROBIAL CONTEXT

START / CONCEPT: CORE SYNTHESIS - Microbiome probiotics extremophiles and microbial context
|
v
EXACT TERMS: CORE SYNTHESIS · Microbiome probiotics extremophiles · microbial context · Mechanism chain · Qualified use · A biofilm
|
v
MECHANISM / ARGUMENT: Read Microbiome probiotics extremophiles and microbial context as a sequence: identify Biofilms microbiome probiotics and extremophiles, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.
|
v
CONSEQUENCE / CONTRAST: Mechanism chain: Biofilms microbiome probiotics and extremophiles Qualified use: State community, host or environmental context before inferring benefit, hazard or treatment response.
|
v
UPSC TRAP / ANSWER-USE: Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context.
|
v
ANSWER-GRABBING FORMULATION: A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion.

SESSION 14 - CORE SYNTHESIS - PLANT PHYSIOLOGY PHOTOSYNTHESIS TRANSPORT HORMONES AND ADAPTATION

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Read Plant physiology photosynthesis transport hormones and adaptation as a sequence: identify Plant physiology transport and stress response, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

Technical definition: Mechanism chain: Plant physiology transport and stress response - Routed taxonomy and organism-form distinctions Qualified use: Trace light, carbon, water, transport tissue, stomatal control and stress strategy.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Plant physiology photosynthesis transport hormones and adaptation as a sequence: identify Plant physiology transport and stress response, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- CORE SYNTHESIS
- Plant physiology photosynthesis transport hormones
- adaptation
- Mechanism chain
- Qualified use
- Stomata

How to use them: Frame the answer through CORE SYNTHESIS; define Plant physiology photosynthesis transport hormones, connect adaptation with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

PLANT PHYSIOLOGY PHOTOSYNTHESIS TRANSPORT HORMONES AND ADAPTATION

01. Plant physiology transport and stress response

|
v

02. Routed taxonomy and organism-form distinctions

STATUS / CATEGORY FIREWALL -> Do not treat bacteria, viruses, fungi, protozoa, helminths and vectors as one category.

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties.

SIMPLE SYSTEM EXAMPLE

Read Plant physiology photosynthesis transport hormones and adaptation as a sequence: identify Plant physiology transport and stress response, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not treat bacteria, viruses, fungi, protozoa, helminths and vectors as one category. This keeps Plant physiology transport and stress response and Routed taxonomy and organism-form distinctions in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies.
- Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties.

EXAMINER CAUTION

- Do not treat bacteria, viruses, fungi, protozoa, helminths and vectors as one category.

EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Trace light, carbon, water, transport tissue, stomatal control and stress strategy.

MINI RECAP

- **Mechanism chain:** Plant physiology transport and stress response -> Routed taxonomy and organism-form distinctions
- **Qualified use:** Trace light, carbon, water, transport tissue, stomatal control and stress strategy.

CLOSING RECALL FLOW - CORE SYNTHESIS - PLANT PHYSIOLOGY PHOTOSYNTHESIS TRANSPORT HORMONES AND ADAPTATION

START / CONCEPT: CORE SYNTHESIS - Plant physiology photosynthesis transport hormones and adaptation

|
v

EXACT TERMS: CORE SYNTHESIS · Plant physiology photosynthesis transport hormones · adaptation · Mechanism chain · Qualified use · Stomata

|
v

MECHANISM / ARGUMENT: Mechanism chain: Plant physiology transport and stress response - Routed taxonomy and organism-form distinctions Qualified use: Trace light, carbon, water, transport tissue, stomatal control and stress strategy.

|
v

CONSEQUENCE / CONTRAST: Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity.

|
v

UPSC TRAP / ANSWER-USE: The decisive boundary is Do not treat bacteria, viruses, fungi, protozoa, helminths and vectors as one category.

|
v

ANSWER-GRABBING FORMULATION: Read Plant physiology photosynthesis transport hormones and adaptation as a sequence: identify Plant physiology transport and stress response, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

SESSION 15 - CORE SYNTHESIS - ROUTED ORGANISMS CELLULOSE INSTITUTIONS PYQS AND EVIDENCE STATUS

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes.

Technical definition: Read Routed organisms cellulose institutions PYQs and evidence status as a sequence: identify Cellulose decomposition route, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Read Routed organisms cellulose institutions PYQs and evidence status as a sequence: identify Cellulose decomposition route, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- CORE SYNTHESIS
- Routed organisms cellulose institutions PYQs
- evidence status
- Mechanism chain
- Qualified use
- Subject

How to use them: Frame the answer through CORE SYNTHESIS; define Routed organisms cellulose institutions PYQs, connect evidence status with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

ROUTED ORGANISMS CELLULOSE INSTITUTIONS PYQS AND EVIDENCE STATUS

01. Cellulose decomposition route

|
v

02. Health institution and evidence-status firewall

STATUS / CATEGORY FIREWALL -> Do not route vaccine platforms, monoclonal antibodies or gene-editing mechanisms into this fundamentals owner.

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs.

SIMPLE SYSTEM EXAMPLE

Read Routed organisms cellulose institutions PYQs and evidence status as a sequence: identify Cellulose decomposition route, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not route vaccine platforms, monoclonal antibodies or gene-editing mechanisms into this fundamentals owner. This keeps Cellulose decomposition route and Health institution and evidence-status firewall in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process.
- ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs.

EXAMINER CAUTION

- Do not route vaccine platforms, monoclonal antibodies or gene-editing mechanisms into this fundamentals owner.

EXAM LINK

- **Prelims:** Keep institution, system, platform, service, process, legal instrument, unit and status in their exact category.
- **Mains:** Route each printed demand to the exact taxonomy, decomposition or institution distinction and stop at the last verified status.

MINI RECAP

- **Mechanism chain:** Cellulose decomposition route -> Health institution and evidence-status firewall
- **Qualified use:** Route each printed demand to the exact taxonomy, decomposition or institution distinction and stop at the last verified status.

COMPLETE BASIC OWNER EVIDENCE BANK

Subject: Science & Technology | **Tier:** Must-Do (foundation) | **GS Paper:** GS-III + Prelims (*primarily Prelims-heavy general science*). **Core area:** Cell biology, human physiology, genetics, immunity, microbiology and plant physiology relevant to UPSC general science. **Grounded in:** NCERT Class IX-X Science; NCERT Class XI-XII Biology; Nobel Prize in Physiology or Medicine 2024 press release

(<https://www.nobelprize.org/prizes/medicine/2024/press-release/>); WHO trachoma validation page (<https://www.who.int/news/item/08-10-2024-elimination-of-trachoma-as-a-public-health-problem-in-india>); WHO India polio transition snapshot (<https://www.who.int/publications/m/item/india-polio-transition-snapshot>); WHO/UNICEF immunization-coverage methodology page (<https://www.who.int/teams/immunization-vaccines-and-biologicals/immunization-analysis-and-insights/global-monitoring/immunization-coverage/who-unicef-estimates-of-national-immunization-coverage>) - verified 16 Jul 2026. **FACT:** = source-grounded | **ANALYSIS:** = analytical inference | **CURRENT:** = current/dated development. *Companion:* advanced/23_General-Science-Biology-and-Physiology.md.

1. Visual foundation

System / unit	Core function	UPSC cue
FACT: Cell	Basic structural and functional unit of life	Organelles, DNA, metabolism
FACT: Digestive system	Breakdown and absorption of food	Enzymes, nutrients, liver-pancreas role
FACT: Circulatory system	Transport of gases, nutrients, hormones and wastes	Heart, blood, hemoglobin
FACT: Respiratory system	Gas exchange and cellular oxygen supply	Lungs vs cellular respiration
FACT: Nervous + endocrine systems	Rapid electrical control + slower chemical coordination	Reflexes, hormones, homeostasis
FACT: Immune system	Recognition and defense against foreign agents	Antigen, antibody, innate/adaptive immunity

DNA → RNA → protein
 | |
 | +--> regulation / gene expression
 +--> inheritance

antigen exposure → immune recognition → response → memory

Core proposition: UPSC biology rewards candidates who distinguish structure, function and regulation instead of memorizing isolated terms.

2. Essential definitions

Term	Exam-ready meaning
FACT: Cell	Basic structural and functional unit of life.
FACT: Organelle	Specialized subcellular structure performing a defined function inside a cell.
FACT: Mitochondrion	Organelle associated with aerobic respiration and ATP generation.
FACT: Gene	Functional unit of heredity located on DNA.
FACT: DNA	Double-stranded nucleic acid storing hereditary information in most organisms.
FACT: RNA	Usually single-stranded nucleic acid involved in expression and regulation of genetic information.
FACT: Hormone	Chemical messenger secreted in small amounts and carried to target tissues.
FACT: Antigen	Any substance that can be recognised and bound by an antibody or T-cell receptor . ANALYSIS: Antigens need not be foreign: self-antigens exist, and immune tolerance normally prevents a response to them - failure of tolerance produces autoimmune disease . An antigen that actually provokes a response is an immunogen .
FACT: Antibody	Specific protein produced by B-lymphocyte lineage that binds an antigen.
FACT: Innate immunity	Inborn, immediate and non-specific defence - physical/chemical barriers (skin, mucus, stomach acid, lysozyme), phagocytes (neutrophils, macrophages), natural killer cells, complement proteins, interferons and the inflammatory and fever responses.
FACT: Adaptive immunity	Specific acquired defense involving memory, B cells and T cells.
FACT: Photosynthesis	Process by which green plants use light energy to synthesize food from carbon dioxide and water.

High-frequency Prelims biology - the compact coverage table.

Area	What must be known
ANALYSIS: Cell structure	Prokaryote (no true nucleus or membrane-bound organelles - bacteria) vs eukaryote . Organelles: nucleus (genetic control), ribosome (protein synthesis), rough/smooth ER (protein and lipid processing), Golgi (packaging and secretion), lysosome ("suicide bag", hydrolytic enzymes), mitochondrion (ATP), chloroplast and cell wall (plants only), vacuole . Membrane transport: diffusion , osmosis , active transport .
ANALYSIS: Cell division	Mitosis - one division, two genetically identical diploid daughter cells, for growth and repair. Meiosis - two divisions, four genetically varied haploid gametes, with crossing over generating variation. Confusing the two is a standard error.
ANALYSIS: Molecular genetics	DNA replication is semi-conservative; transcription makes mRNA; translation at the ribosome uses tRNA and the genetic code (triplet codons) to build protein. Mutations (point, frameshift, chromosomal) are the raw material of variation; sickle-cell anaemia and thalassaemia are standard Indian examples.
ANALYSIS: Mendelian inheritance	Laws of dominance, segregation and independent assortment; monohybrid 3:1 and dihybrid 9:3:3:1 ratios. Sex determination in humans is XX/XY (the sperm determines sex). Sex-linked conditions such as haemophilia and red-green colour blindness are X-linked recessive, hence commoner in males.

Area	What must be known
ANALYSIS: Blood	ABO groups with A, B, AB (universal recipient) and O (universal donor) ; Rh factor - Rh-negative mothers carrying an Rh-positive foetus risk erythroblastosis foetalis . Blood components: plasma, RBCs (haemoglobin, no nucleus in humans), WBCs, platelets (clotting).
ANALYSIS: Human systems	Circulatory - four-chambered heart, double circulation, arteries carry blood away (pulmonary artery is the deoxygenated exception). Respiratory - gas exchange in alveoli; haemoglobin binds oxygen but binds carbon monoxide far more strongly , which is why CO poisoning is lethal. Excretory - the nephron filters (glomerulus), reabsorbs and secretes; dialysis substitutes for failed kidneys. Nervous - CNS and peripheral; neurons transmit via electrical impulses and neurotransmitters; the medulla controls involuntary functions; reflex arcs bypass the brain. Digestive - enzymes act at specific pH (salivary amylase, gastric pepsin in acid, pancreatic and intestinal enzymes in alkaline medium); bile emulsifies fat.
ANALYSIS: Endocrine system	Pituitary (master gland: growth hormone), thyroid (thyroxine; iodine deficiency -> goitre), parathyroid (calcium), adrenal (adrenaline, cortisol), pancreas (insulin and glucagon - diabetes mellitus), gonads.
ANALYSIS: Nutrition and deficiency	Fat-soluble vitamins A, D, E, K ; water-soluble B-complex and C . Deficiencies: A -> night blindness/xerophthalmia ; D -> rickets and osteomalacia ; C -> scurvy ; B1 -> beriberi ; B3 -> pellagra ; B12/folate -> anaemia ; K -> clotting defects ; iron -> anaemia ; iodine -> goitre and cretinism . Protein-energy malnutrition: kwashiorkor (protein deficit, oedema) and marasmus (overall energy deficit, wasting).
ANALYSIS: Diseases and pathogens	Bacterial - TB, cholera, typhoid, tetanus, leprosy. Viral - measles, polio, dengue, chikungunya, rabies, hepatitis, COVID-19, influenza. Protozoan - malaria (Plasmodium, female Anopheles) , kala-azar (Leishmania, sandfly), amoebiasis. Fungal - ringworm, candidiasis. Helminthic - filariasis (Culex), ascariasis. Vectors matter : <i>Aedes aegypti</i> -> dengue/chikungunya/Zika; <i>Anopheles</i> -> malaria; <i>Culex</i> -> filariasis/Japanese encephalitis. Zoonoses (rabies, Nipah, avian influenza, brucellosis) justify the One Health approach.
ANALYSIS: Communicable vs non-communicable	NCDs - cardiovascular disease, diabetes, cancers, chronic respiratory disease, mental illness - now dominate India's disease burden and are driven by tobacco, alcohol, diet, physical inactivity and air pollution. India's double burden of persisting infections plus rising NCDs is the standard framing.
ANALYSIS: Antibiotics vs antivirals; AMR	Antibiotics act on bacteria only - they do nothing against viruses. Antimicrobial resistance arises from over-prescription, incomplete courses, over-the-counter sales, use in animal husbandry and antibiotic residues in the environment; it is a selection process, not "the body becoming resistant." India's National Action Plan on AMR and Schedule H1 prescription controls are the policy responses.
ANALYSIS: Human microbiome	Trillions of gut and skin microbes involved in digestion, vitamin synthesis (K, some B), immune development and pathogen exclusion; disturbed by antibiotics; probiotics add live beneficial microbes and prebiotics feed them.

Area	What must be known
ANALYSIS: Plant physiology	Photosynthesis (light and dark reactions, chlorophyll, stomata), transpiration and the transpiration pull driving water movement in xylem ; phloem translocates food. Plant hormones: auxin (elongation, apical dominance), gibberellin (stem growth), cytokinin (cell division), abscisic acid (stress and stomatal closure), ethylene (fruit ripening). C3, C4 and CAM pathways explain drought adaptation.

3. Mechanism / how it works

How adaptive immune response develops (without entering vaccine-platform details)

1. FACT: An antigen enters the body and is recognised as **non-self**. ANALYSIS: Recognition of **self** antigens is normally suppressed by **immune tolerance**; when tolerance fails, autoimmune disease results.
2. FACT: Antigen-presenting cells help activate lymphocytes.
3. FACT: B cells differentiate into plasma cells that produce antibodies; T cells coordinate or directly attack depending on subtype.
4. FACT: Antibodies bind specific antigens and help neutralize or mark them for destruction.
5. FACT: Some lymphocytes become memory cells, enabling a faster response on future exposure.
6. ANALYSIS: This is the conceptual immune base; vaccine platforms, adjuvants and regulatory approval belong in 15_Vaccines-Monoclonal-Antibodies-and-Biopharma.md, not here.

4. Institutions and programmes

- FACT: **DBT (Department of Biotechnology)**: major biotechnology-policy and programme institution for applied life sciences.
- FACT: **CSIR-CCMB**: cellular and molecular biology research ecosystem.
- FACT: **CSIR-IGIB**: genomics and integrative biology research.
- FACT: **ICMR**: India's apex body for **biomedical research** - it formulates research policy, runs institutes and issues ethical guidelines. ANALYSIS: It is **not** the operational disease-surveillance agency: **NCDC** and the **IDSP/IHIP** network under **MoHFW**, together with State health departments, perform surveillance and outbreak response, while **CDSCO** approves drugs and vaccines. Collapsing ICMR, NCDC, MoHFW and CDSCO into one "public-health ecosystem" is a common institutional error.
- ANALYSIS: Foundations here support later applied discussions in 13_Biotechnology-Fundamentals-and-DBT-Missions.md, 14_Genetic-Engineering-GM-Crops-and-CRISPR.md and 15_Vaccines-Monoclonal-Antibodies-and-Biopharma.md.

5. Indian applications, examples and limitations

- FACT: Cell biology and genetics support diagnostics, disease surveillance, cancer research and reproductive health.
- FACT: Human physiology is the base for public-health questions on nutrition, blood, respiration, endocrine disorders and stress responses.
- FACT: Immunology underpins vaccination, serology and disease-control strategies, even when the applied technology is discussed elsewhere.
- FACT: Microbiology matters for infectious disease, fermentation, agriculture and biotechnology.
- FACT: Plant physiology remains relevant for crop productivity because photosynthesis, respiration and nutrient use directly shape yield.
- ANALYSIS: **Limitation / pitfall 1**: DNA/RNA basics should not be confused with gene-editing mechanism details; cross-reference Topic 14 instead of duplicating it.
- ANALYSIS: **Limitation / pitfall 2**: immunity basics are not the same as vaccine technology, cold chains or regulatory approval; those belong in Topic 15.

- ANALYSIS: **Limitation / pitfall 3:** virus, bacteria and fungi are biologically different classes; treating them as one bucket causes many Prelims errors.

6. Must-Know Facts for Prelims

- FACT: Prokaryotic cells lack a true membrane-bound nucleus; eukaryotic cells possess one.
- FACT: Mitochondria are linked with cellular respiration; chloroplasts with photosynthesis.
- FACT: The small intestine is the principal site of digestion completion and nutrient absorption.
- FACT: Hemoglobin carries oxygen; blood plasma carries dissolved substances and proteins.
- FACT: The nervous system acts rapidly through impulses; the endocrine system acts more slowly through hormones.
- FACT: DNA contains deoxyribose sugar and thymine; RNA contains ribose sugar and uracil.
- FACT: Bacteria are living cellular organisms; viruses are acellular particles that require host cells for replication.
- FACT: Fungi are eukaryotic organisms and are neither plants nor bacteria.
- FACT: Innate immunity is non-specific; adaptive immunity is specific and can develop memory.
- FACT: Photosynthesis and respiration are related but not identical processes; plants respire all the time.

7. UPSC traps

- WRONG: **All cells in the body perform the same function because they contain the same DNA.** -> Differential gene expression creates functional specialization.
- WRONG: **DNA and RNA differ only in name.** -> They differ in sugar, one nitrogen base, typical structure and primary role.
- WRONG: **Antibodies and antigens are the same thing.** -> Antigen triggers the response; antibody is part of the response.
- WRONG: **Virus and bacteria can be treated identically in biology questions.** -> One is acellular and host-dependent; the other is cellular and independently living.
- WRONG: **Hormones and enzymes are interchangeable terms.** -> Hormones are signaling molecules; enzymes are biological catalysts.
- WRONG: **Plants only photosynthesize and do not respire.** -> Plants respire continuously; photosynthesis is light-dependent.

8. CURRENT: Current anchor

- CURRENT: **07 Oct 2024 | Nobel Prize in Physiology or Medicine.** The prize recognized microRNA and post-transcriptional gene regulation, reinforcing the importance of RNA beyond the old "DNA alone stores information" simplification.
- CURRENT: **08 Oct 2024 | WHO.** India was validated as having eliminated trachoma as a public-health problem - a useful anchor for disease biology, hygiene and public-health application of microbiology.
- CURRENT: **Apr 2024 | WHO India polio transition snapshot.** The snapshot emphasizes sustaining immunization, surveillance and emergency-response systems after eradication success.
- CURRENT: **Data reported till 21 Jun 2026 | WHO/UNICEF immunization-estimate methodology page.** Annual coverage estimates remain a live reminder that biology and physiology knowledge directly enters public-health monitoring.

Current as of 16 Jul 2026; verify for later updates.

10. Mains framework / angles

- ANALYSIS: Use biology fundamentals to explain public health, nutrition, genomics, disease surveillance and human capital.
- ANALYSIS: When writing on genetics, perform the mandatory distinction between basic molecular biology here and applied genetic engineering in Topic 14.
- ANALYSIS: When writing on immunology, keep this file to antigen-antibody and immunity logic; move vaccine platforms and regulation to Topic 15.
- ANALYSIS: Add plant physiology where agriculture-productivity or climate-stress questions need a scientific base.

Answer thesis: General biology and physiology matter in GS-III because they connect molecular and organ-level science to public health, disease control, nutrition, biotechnology and agricultural productivity.

11. Probable questions

- ANALYSIS: **Practice Prelims:** Which of the following correctly distinguishes DNA, RNA, antigen, antibody and hormone?
- ANALYSIS: **Practice Mains (10 marks):** Explain how the nervous and endocrine systems coordinate to maintain homeostasis in the human body. *Answer in 150 words.*
- ANALYSIS: **Practice Mains (15 marks):** Discuss the relevance of basic genetics, immunology and microbiology for India's public-health and biotechnology agenda.

12. Study links

- FACT: Advanced companion: advanced/23_General-Science-Biology-and-Physiology.md.
- FACT: 21_General-Science-Physics-Fundamentals.md - imaging, radiation and instrumentation frequently intersect with biology.
- FACT: 22_General-Science-Chemistry-Fundamentals.md - biomolecules begin there as chemistry and deepen here as life processes.
- FACT: 13_Biotechnology-Fundamentals-and-DBT-Missions.md - applied biotech architecture.
- FACT: 14_Genetic-Engineering-GM-Crops-and-CRISPR.md and 15_Vaccines-Monoclonal-Antibodies-and-Biopharma.md - the applied extensions explicitly separated from this fundamentals file.

Core answer architecture - biology, physiology and health/ecology reasoning

Thesis choice. Biology questions require mechanism across levels-cell, molecule, organism, population and system; a named organ or microbe is not an explanation unless its function and boundary are shown.

10-mark spine. Define the structure/process; trace the biological mechanism; give one health/agriculture/ecology application; add a misconception or variation.

15/20-mark spine. Use **cellular/molecular basis -> organ/system or ecological outcome -> public-health/agriculture application -> population variation, resistance, ethics or access -> measured conclusion.**

Evidence units.

- **Claim:** gene information becomes phenotype through regulated expression -> **DNA transcription to RNA and translation to protein** -> links genes to enzymes, receptors, immunity and biotechnology -> **qualification:** a gene association does not prove one-gene causation for complex traits or disease.
- **Claim:** immune protection has different arms and time profiles -> **B cells produce antibodies while T cells coordinate/carry out cellular functions; vaccines train active immunity** -> supports reasoning about infection, vaccination and therapeutic antibodies -> **qualification:** an immune response varies by pathogen, host and exposure; antibiotics do not treat viral disease.
- **Claim:** microbial control is an evolutionary and system problem -> **selection under antimicrobial exposure, biofilms and host/environmental transmission** -> explains why resistance, hygiene and stewardship must be combined -> **qualification:** resistance resides in microbial populations; neither a person nor a whole species becomes uniformly "immune" through one exposure.

Verdict. Use mechanism to guide prevention and care while avoiding biological determinism, overgeneralisation from one species and unsupported medical claims.

Historical Mains Core route - cellulose decomposition (2022)

Cellulose is a plant structural polysaccharide. Decomposer fungi and bacteria secrete **cellulases** that hydrolyse it into smaller sugars; microbial metabolism then returns carbon through soil/food-web processes. **Significance:** this supports litter decomposition, composting and biomass-to-biofuel/bioprocess possibilities. **Qualification:** lignin-rich material, moisture, temperature, oxygen and microbial community constrain rate; natural decomposition is not the same as an industrially viable biofuel process.

CLOSING RECALL FLOW - CORE SYNTHESIS - ROUTED ORGANISMS CELLULOSE INSTITUTIONS PYQS AND EVIDENCE STATUS

START / CONCEPT: CORE SYNTHESIS - Routed organisms cellulose institutions PYQs and evidence status

|
v

EXACT TERMS: CORE SYNTHESIS · Routed organisms cellulose institutions PYQs · evidence status · Mechanism chain · Qualified use · Subject

|
v

MECHANISM / ARGUMENT: Mechanism chain: Cellulose decomposition route - Health institution and evidence-status firewall Qualified use: Route each printed demand to the exact taxonomy, decomposition or institution distinction and stop at the last verified status.

|
v

CONSEQUENCE / CONTRAST: Biology questions require mechanism across levels-cell, molecule, organism, population and system; a named organ or microbe is not an explanation unless its function and boundary are shown.

|
v

UPSC TRAP / ANSWER-USE: FACT: DBT (Department of Biotechnology): major biotechnology-policy and programme institution for applied life sciences. FACT: CSIR-CCMB: cellular and molecular biology research ecosystem. FACT: CSIR-IGIB: genomics and integrative biology research. FACT: ICMR: India's apex body for biomedical research - it formulates research policy, runs institutes and issues ethical guidelines. ANALYSIS: It is not the operational disease-surveillance agency: NCDC and the IDSP/IHIP network under MoHFW, together with State health departments, perform surveillance and outbreak response, while CDSCO approves drugs and vaccines.

|
v

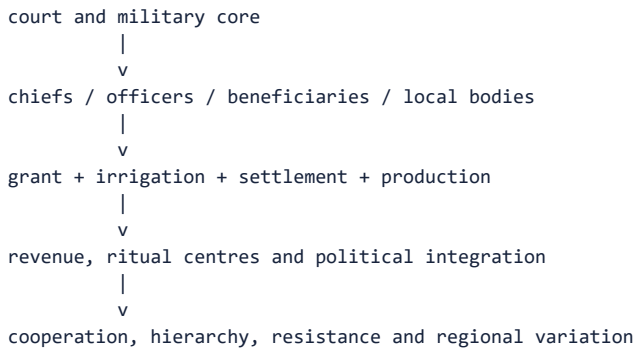
ANSWER-GRABBING FORMULATION: Read Routed organisms cellulose institutions PYQs and evidence status as a sequence: identify Cellulose decomposition route, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

CLOSING SEMANTIC LEDGER A - WATER, GRANT FORMS, INTERMEDIARIES AND STATE DEBATE

Why this supplement is necessary

[ANALYSIS] A complete account cannot move directly from dynastic war to temples. It must show the material and intermediary mechanisms that connected a court to uneven local societies.

Missing bridge	Evidence-led reconstruction	Limit
Irrigation	Tanks, wells and channels supported cultivation in rainfall-variable zones; donors, local bodies and institutions could share maintenance	An inscription naming a work does not prove uniform state irrigation
Grant forms	<i>Brahmadeya</i> and <i>agrahara</i> primarily concern Brahmana beneficiaries; <i>devadana</i> assigns resources to a deity or temple	Formula and actual enforcement must be separated
Intermediary power	Princes, officers, tributary chiefs, military retainers, village bodies and beneficiaries connected core courts to localities	A campaign zone is not automatically an annexed province
Production society	Peasants, pastoralists, artisans and service groups sustained agrarian and temple networks	Elite records create severe source silence
Gendered patronage	Queen Lokamahadevi is associated with Virupaksha and Queen Trailokyamahadevi with Mallikarjuna at Pattadakal	Elite female patronage does not establish general equality



State-formation debate

Model	What it illuminates	Why it is insufficient alone
Indian-feudalism interpretation	Fiscal immunities, landed intermediaries and possible peasant dependence	Grants varied; royal capacity, towns and exchange did not vanish uniformly
Segmentary-state heuristic	Stronger core, graded authority and ritual claims beyond direct control	It was developed most fully for later Chola evidence and cannot be pasted onto every Pallava-Chalukya phase
Integrative/processual approach	Grants, cults, alliances and local institutions as polity-building mechanisms	Integration remained unequal and conflictual
Peasantization/agrarian-expansion approach	Incorporation of forest, pastoral and cultivating populations	Change was not one-way or identical in every ecology

Qualified verdict: Pallava and Chalukya states combined coercion, tribute, redistribution, intermediary negotiation and ritual-cultural integration.

CLOSING SEMANTIC LEDGER B - NEGOTIATED CULTURE, LANGUAGE AND BOUNDARY CONTROL

Brahmanization was negotiated and uneven

- [FACT] Grants record Brahmana migration and beneficiary networks, but they are not complete population maps.

- [ANALYSIS] *Sanskritization* identifies some adoption of high-status practices;

localization and vernacularization show how received forms were changed in Tamil, Kannada and Telugu-region settings.

- [ANALYSIS] Puranic identifications could incorporate local deities without erasing local names, myths, festivals or constituencies.

- [LIMIT] Varna idioms interacted with occupational and locality-based jatis; the peninsula did not become a copy of a northern normative text.

- [LIMIT] Bhakti could broaden devotional access and criticize hierarchy, yet temple institutions could reproduce rank. It was not solely an anti-caste protest.

Language and script matrix

Context	Exam-safe formulation
Pallava chancery	Early Prakrit and then Sanskrit charters; later Sanskrit eulogy often coexisted with Tamil documentary content
Badami Chalukya realm	Sanskrit court poetry coexisted with expanding Kannada epigraphic expression
Vengi	Sanskrit prestige interacted with regional documentary practice; major Telugu literary florescence belongs substantially to later centuries
Literacy	Epigraphic production proves institutional literacy, not mass literacy

Correct eastern and transition boundary

- [FACT] Pulakeshin II's brother Kubja Vishnuvardhana founded the Eastern Chalukya line in Vengi in the early seventh century.
- [FACT] The branch continued into the eleventh century. Rajaraja I intervened in Vengi succession politics and backed restoration; this must not be rewritten as a simple conquest ending the dynasty in AD 999.
- [FACT] Rashtrakuta displacement of Kirtivarman II is conventionally placed around AD 753. It ends Badami Chalukya rule, not every Chalukya line.
- [BOUNDARY] Full Rashtrakuta history and later Chola-Vengi integration remain outside Topic 23.

Final source-control checklist

1. Court eulogy is evidence of a claim, not an audited campaign map.
2. A grant formula is not proof that every listed right was uniformly exercised.
3. Monument style is not a rigid ethnic identity.
4. Royal patronage does not erase artisans, queens, local donors or institutions.
5. Modern linguistic-state boundaries must not be projected backward.
6. Contact with Sri Lanka or Southeast Asia is not proof of a colonial empire.

SCIENCE AND TECHNOLOGY DEEP-REVIEW CORE CONTROL

- **Must remember:** Biology and physiology connect cell structure, metabolism, genetics, evolution, microbes, plant/animal systems, immunity, nervous/endocrine control, circulation, respiration, digestion, excretion and reproduction.
- **Close distinction:** Gene is not allele or genome, genotype is not phenotype, pathogen is not disease, infection is not symptom severity, antibody is not antibiotic, hormone is not enzyme, and association is not clinical causation.
- **Mechanism / status / evidence limit:** Name organism/cell/tissue/organ, mechanism, pathway, evidence level and normal physiological range only from a source; distinguish prevention, diagnosis, treatment, approval and population outcome without giving medical advice.

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Cell organisation and organelle boundary?

A. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. B. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. C. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. D. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context.

Answer: A. Explanation: A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation,

the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of Cell organisation and organelle boundary?

A. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. B. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. C. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. D. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context.

Answer: B. Explanation: A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses Cell organisation and organelle boundary without changing its institution, unit or status?

A. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. B. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. C. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. D. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages.

Answer: C. Explanation: A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about Cell organisation and organelle boundary?

A. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. B. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. C. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. D. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole.

Answer: D. Explanation: A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies Biomolecules and enzyme specificity?

A. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. B. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. C. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. D. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages.

Answer: A. Explanation: Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of Biomolecules and enzyme specificity?

A. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. B. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. C. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. D. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns.

Answer: B. Explanation: Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses Biomolecules and enzyme specificity without changing its institution, unit or status?

A. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. B. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. C. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. D. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.

Answer: C. Explanation: Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about Biomolecules and enzyme specificity?

A. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. B. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. C. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. D. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function.

Answer: D. Explanation: Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies Membrane transport and water balance?

A. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. B. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. C. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. D. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages.

Answer: A. Explanation: Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of Membrane transport and water balance?

A. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. B. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. C. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. D. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages.

Answer: B. Explanation: Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses Membrane transport and water balance without changing its institution, unit or status?

A. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. B. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. C. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. D. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.

Answer: C. Explanation: Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about Membrane transport and water balance?

A. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. B. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. C. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. D. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context.

Answer: D. Explanation: Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies Mitosis meiosis and variation?

A. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. B. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. C. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. D. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.

Answer: A. Explanation: Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of Mitosis meiosis and variation?

A. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. B. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. C. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. D. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.

Answer: B. Explanation: Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses Mitosis meiosis and variation without changing its institution, unit or status?

A. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. B. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. C. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. D. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions.

Answer: C. Explanation: Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about Mitosis meiosis and variation?

A. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. B. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. C. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. D. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell.

Answer: D. Explanation: Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies DNA RNA gene expression and microRNA?

A. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. B. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. C. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. D. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.

Answer: A. Explanation: DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of DNA RNA gene expression and microRNA?

A. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. B. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. C. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. D. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions.

Answer: B. Explanation: DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses DNA RNA gene expression and microRNA without changing its institution, unit or status?

A. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. B. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. C. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. D. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous.

Answer: C. Explanation: DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about DNA RNA gene expression and microRNA?

A. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. B. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. C. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. D. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages.

Answer: D. Explanation: DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Inheritance mutation and sex linkage?

A. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. B. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. C. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. D. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.

Answer: A. Explanation: A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Inheritance mutation and sex linkage?

A. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. B. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. C. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. D. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys.

Answer: B. Explanation: A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Inheritance mutation and sex linkage without changing its institution, unit or status?

A. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. B. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. C. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. D. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous.

Answer: C. Explanation: A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Inheritance mutation and sex linkage?

A. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. B. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. C. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. D. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns.

Answer: D. Explanation: A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies Digestion absorption and nutrition?

A. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. B. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. C. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. D. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous.

Answer: A. Explanation: Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of Digestion absorption and nutrition?

A. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. B. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. C. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. D. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems.

Answer: B. Explanation: Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses Digestion absorption and nutrition without changing its institution, unit or status?

A. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. B. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. C. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. D. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems.

Answer: C. Explanation: Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about Digestion absorption and nutrition?

A. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. B. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. C. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. D. Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.

Answer: D. Explanation: Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies Blood heart and double circulation?

A. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. B. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. C. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. D. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys.

Answer: A. Explanation: Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of Blood heart and double circulation?

A. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. B. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. C. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. D. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms.

Answer: B. Explanation: Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses Blood heart and double circulation without changing its institution, unit or status?

A. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. B. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. C. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. D. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms.

Answer: C. Explanation: Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about Blood heart and double circulation?

A. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. B. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. C. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. D. Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions.

Answer: D. Explanation: Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies Respiration gas exchange and cellular energy?

A. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. B. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. C. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. D. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms.

Answer: A. Explanation: Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of Respiration gas exchange and cellular energy?

A. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. B. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. C. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. D. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems.

Answer: B. Explanation: Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses Respiration gas exchange and cellular energy without changing its institution, unit or status?

A. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. B. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. C. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. D. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone.

Answer: C. Explanation: Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about Respiration gas exchange and cellular energy?

A. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. B. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. C. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. D. Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous.

Answer: D. Explanation: Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies Kidney nephron and homeostasis?

A. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. B. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. C. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. D. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone.

Answer: A. Explanation: The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of Kidney nephron and homeostasis?

A. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. B. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. C. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate

calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. D. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself.

Answer: B. Explanation: The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses Kidney nephron and homeostasis without changing its institution, unit or status?

A. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. B. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. C. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. D. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body.

Answer: C. Explanation: The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about Kidney nephron and homeostasis?

A. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. B. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. C. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. D. The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys.

Answer: D. Explanation: The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies Nervous endocrine and feedback control?

A. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. B. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. C. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. D. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone.

Answer: A. Explanation: Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of Nervous endocrine and feedback control?

A. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. B. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. C. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. D. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body.

Answer: B. Explanation: Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses Nervous endocrine and feedback control without changing its institution, unit or status?

A. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. B. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. C. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. D. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body.

Answer: C. Explanation: Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about Nervous endocrine and feedback control?

A. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. B. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. C. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. D. Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems.

Answer: D. Explanation: Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies Hormones metabolism and internal stability?

A. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. B. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. C. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. D. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself.

Answer: A. Explanation: Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of Hormones metabolism and internal stability?

A. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. B. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. C. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. D. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body.

Answer: B. Explanation: Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses Hormones metabolism and internal stability without changing its institution, unit or status?

A. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. B. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. C. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. D. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies.

Answer: C. Explanation: Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about Hormones metabolism and internal stability?

A. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. B. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. C. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, froghopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. D. Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms.

Answer: D. Explanation: Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies Innate adaptive and tolerance architecture?

A. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. B. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. C. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. D. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself.

Answer: A. Explanation: Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of Innate adaptive and tolerance architecture?

A. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. B. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. C. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. D. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context.

Answer: B. Explanation: Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses Innate adaptive and tolerance architecture without changing its institution, unit or status?

A. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. B. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. C. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. D. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context.

Answer: C. Explanation: Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about Innate adaptive and tolerance architecture?

A. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. B. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. C. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. D. Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone.

Answer: D. Explanation: Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies Pathogen transmission vector and zoonosis map?

A. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. B. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. C. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. D. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context.

Answer: A. Explanation: Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of Pathogen transmission vector and zoonosis map?

A. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. B. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. C. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. D. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context.

Answer: B. Explanation: Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses Pathogen transmission vector and zoonosis map without changing its institution, unit or status?

A. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. B. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. C. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. D. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties.

Answer: C. Explanation: Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about Pathogen transmission vector and zoonosis map?

A. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. B. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. C. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. D. Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself.

Answer: D. Explanation: Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies Antibiotic antiviral and AMR boundary?

A. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. B. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. C. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. D. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies.

Answer: A. Explanation: Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of Antibiotic antiviral and AMR boundary?

A. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. B. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. C. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. D. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies.

Answer: B. Explanation: Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses Antibiotic antiviral and AMR boundary without changing its institution, unit or status?

A. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, froghopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. B. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. C. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. D. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process.

Answer: C. Explanation: Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about Antibiotic antiviral and AMR boundary?

A. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. B. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. C. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. D. Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body.

Answer: D. Explanation: Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies Biofilms microbiome probiotics and extremophiles?

A. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. B. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. C. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. D. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process.

Answer: A. Explanation: A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of Biofilms microbiome probiotics and extremophiles?

A. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. B. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. C. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. D. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs.

Answer: B. Explanation: A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses Biofilms microbiome probiotics and extremophiles without changing its institution, unit or status?

A. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. B. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. C. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. D. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs.

Answer: C. Explanation: A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about Biofilms microbiome probiotics and extremophiles?

A. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. B. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. C. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. D. A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context.

Answer: D. Explanation: A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies Plant physiology transport and stress response?

A. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C₃, C₄ and CAM are different carbon-fixation strategies. B. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. C. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. D. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process.

Answer: A. Explanation: Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C₃, C₄ and CAM are different carbon-fixation strategies. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of Plant physiology transport and stress response?

A. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. B. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C₃, C₄ and CAM are different carbon-fixation strategies. C. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. D. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process.

Answer: B. Explanation: Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C₃, C₄ and CAM are different carbon-fixation strategies. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses Plant physiology transport and stress response without changing its institution, unit or status?

A. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. B. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. C. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C₃, C₄ and CAM are different carbon-fixation strategies. D. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs.

Answer: C. Explanation: Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C₃, C₄ and CAM are different carbon-fixation strategies. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about Plant physiology transport and stress response?

A. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. B. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. C. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. D. Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C₃, C₄ and CAM are different carbon-fixation strategies.

Answer: D. Explanation: Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C₃, C₄ and CAM are different carbon-fixation strategies. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Routed taxonomy and organism-form distinctions?

A. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. B. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. C. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. D. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process.

Answer: A. Explanation: Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Routed taxonomy and organism-form distinctions?

A. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. B. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. C. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. D. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function.

Answer: B. Explanation: Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Routed taxonomy and organism-form distinctions without changing its institution, unit or status?

A. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. B. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. C. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, froghopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. D. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function.

Answer: C. Explanation: Groundnut, horse-gram and soybean are routed pea-family examples; cicada, froghopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Routed taxonomy and organism-form distinctions?

A. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. B. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. C. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. D. Groundnut, horse-gram and soybean are routed pea-family examples; cicada, froghopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties.

Answer: D. Explanation: Groundnut, horse-gram and soybean are routed pea-family examples; cicada, froghopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies Cellulose decomposition route?

A. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. B. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. C. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. D. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole.

Answer: A. Explanation: Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of Cellulose decomposition route?

A. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. B. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. C. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. D. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole.

Answer: B. Explanation: Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses Cellulose decomposition route without changing its institution, unit or status?

A. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. B. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. C. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. D. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context.

Answer: C. Explanation: Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about Cellulose decomposition route?

A. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. B. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. C. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. D. Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process.

Answer: D. Explanation: Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Health institution and evidence-status firewall?

A. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. B. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. C. A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. D. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context.

Answer: A. Explanation: ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Health institution and evidence-status firewall?

A. Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. B. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. C. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. D. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell.

Answer: B. Explanation: ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Health institution and evidence-status firewall without changing its institution, unit or status?

A. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. B. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. C. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. D. Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context.

Answer: C. Explanation: ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Health institution and evidence-status firewall?

A. A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. B. Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. C. DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. D. ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs.

Answer: D. Explanation: ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Audited ledgers route thirteen historical Prelims demands from 2018-2023, six Prelims demands from 2024-2025 and the 2022 GS-III cellulose-decomposition demand to this owner. Three representative cards preserve historical objective, recent objective and Mains routes without reproducing answer keys. Vaccine-platform, monoclonal-antibody and gene-editing demands remain with Topics 15 and 14.

PYQ DEMAND CARD 1 - 2018-2023 Prelims GS-I

Demand: Assess routed distinctions on plant adaptations, cell structure, pathogens, artificial culture, viral genomes, ACE2, biofilms, probiotics, B and T cells, mushrooms and extremophiles.

Status: Representative historical objective card covering thirteen routed demands; official keys are unavailable locally, so no option, answer letter or inferred key is supplied.

Model solution: Cell organisation and organelle boundary: A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. **Innate adaptive and tolerance architecture:** Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. **Pathogen transmission vector and zoonosis map:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Antibiotic antiviral and AMR boundary:** Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. **Biofilms microbiome probiotics and extremophiles:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Routed taxonomy and organism-form distinctions:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat "PYQ DEMAND CARD 1 - 2018-2023 Prelims GS-I" as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Cell organisation and organelle boundary: A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. **Innate adaptive and tolerance architecture:** Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and

peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone.

Pathogen transmission vector and zoonosis map: Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself.

Antibiotic antiviral and AMR boundary: Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. **Biofilms microbiome probiotics and extremophiles:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Routed taxonomy and organism-form distinctions:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Assess routed distinctions on plant adaptations, cell structure, pathogens, artificial culture, viral genomes, ACE2, biofilms, probiotics, B and T cells, mushrooms and extremophiles. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Representative historical objective card covering thirteen routed demands; official keys are unavailable locally, so no option, answer letter or inferred key is supplied. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Cell organisation and organelle boundary: A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. **Innate adaptive and tolerance architecture:** Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. **Pathogen transmission vector and zoonosis map:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Antibiotic antiviral and AMR boundary:** Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. **Biofilms microbiome probiotics and extremophiles:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the

human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Routed taxonomy and organism-form distinctions:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For "PYQ DEMAND CARD 1 - 2018-2023 Prelims GS-I", explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

PYQ DEMAND CARD 2 - 2024-2025 Prelims GS-I

Demand: Assess routed organism-family, insect, poisonous-species, nitric-oxide, plant-form and virus-host distinctions.

Status: Representative recent objective card covering six routed demands; official Set-A keys exist locally but no option, answer letter or objective key is reproduced.

Model solution: Hormones metabolism and internal stability: Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. **Pathogen transmission vector and zoonosis map:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Routed taxonomy and organism-form distinctions:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat "PYQ DEMAND CARD 2 - 2024-2025 Prelims GS-I" as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Hormones metabolism and internal stability: Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. **Pathogen transmission vector and zoonosis map:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Routed taxonomy and organism-form distinctions:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The solution

therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Assess routed organism-family, insect, poisonous-species, nitric-oxide, plant-form and virus-host distinctions. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Representative recent objective card covering six routed demands; official Set-A keys exist locally but no option, answer letter or objective key is reproduced. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Hormones metabolism and internal stability: Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. **Pathogen transmission vector and zoonosis map:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Routed taxonomy and organism-form distinctions:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For "PYQ DEMAND CARD 2 - 2024-2025 Prelims GS-I", explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

PYQ DEMAND CARD 3 - 2022 GS-III

Demand: Discuss the natural processes through which cellulose undergoes decomposition on the Earth's surface.

Status: Verified routed Mains demand at 10 marks and 150 words; the response supplies a mechanism-first answer route and qualification, not an official model answer.

Model solution: Biomolecules and enzyme specificity: Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. **Biofilms microbiome probiotics and extremophiles:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Cellulose decomposition route:** Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon

through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: The directive **answer** requires a direct position on "PYQ DEMAND CARD 3 - 2022 GS-III", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Biomolecules and enzyme specificity: Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. **Biofilms microbiome probiotics and extremophiles:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Cellulose decomposition route:** Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Discuss the natural processes through which cellulose undergoes decomposition on the Earth's surface. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Verified routed Mains demand at 10 marks and 150 words; the response supplies a mechanism-first answer route and qualification, not an official model answer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Biomolecules and enzyme specificity: Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. **Biofilms microbiome probiotics and extremophiles:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Cellulose decomposition route:** Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor,

platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "PYQ DEMAND CARD 3 - 2022 GS-III", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish cell types, organelles, biomolecules and membrane-transport mechanisms. Answer in about 150 words.

Model thesis: Claim: Cell organisation and organelle boundary. **Named evidence/example:** A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biomolecules and enzyme specificity. **Named evidence/example:** Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Membrane transport and water balance. **Named evidence/example:** Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole.
- Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function.
- Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context.

Qualified conclusion: Claim: Cell organisation and organelle boundary. **Named evidence/example:** A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall,

chloroplasts and a large vacuole. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biomolecules and enzyme specificity. **Named evidence/example:** Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Membrane transport and water balance. **Named evidence/example:** Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **answer** requires a direct position on "Distinguish cell types, organelles, biomolecules and membrane-transport mechanisms. Answer in...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Cell organisation and organelle boundary. **Named evidence/example:** A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biomolecules and enzyme specificity. **Named evidence/example:** Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Membrane transport and water balance. **Named evidence/example:** Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

2. **Claim and named evidence:** Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

3. **Claim and named evidence:** Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Cell organisation and organelle boundary. **Named evidence/example:** A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biomolecules and enzyme specificity. **Named evidence/example:** Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Membrane transport and water balance. **Named evidence/example:** Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Distinguish cell types, organelles, biomolecules and membrane-transport mechanisms. Answer in...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain how cell division, gene expression, inheritance and mutation generate biological continuity and variation. Answer in about 150 words.

Model thesis: Claim: Mitosis meiosis and variation. **Named evidence/example:** Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DNA RNA gene expression and microRNA. **Named evidence/example:** DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Inheritance mutation and sex linkage. **Named evidence/example:** A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell.
- DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages.
- A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns.

Qualified conclusion: Claim: Mitosis meiosis and variation. **Named evidence/example:** Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DNA RNA gene expression and microRNA. **Named evidence/example:** DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution,

system category, legal character and development-to-deployment status boundary. **Claim:** Inheritance mutation and sex linkage. **Named evidence/example:** A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **explain** requires a direct position on "Explain how cell division, gene expression, inheritance and mutation generate biological...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Mitosis meiosis and variation. **Named evidence/example:** Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DNA RNA gene expression and microRNA. **Named evidence/example:** DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Inheritance mutation and sex linkage. **Named evidence/example:** A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing

benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Mitosis meiosis and variation. **Named evidence/example:** Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DNA RNA gene expression and microRNA. **Named evidence/example:** DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Inheritance mutation and sex linkage. **Named evidence/example:** A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Explain how cell division, gene expression, inheritance and mutation generate biological...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Analyse digestion, circulation, respiration and excretion as an integrated physiological system. Answer in about 250 words.

Model thesis: Claim: Digestion absorption and nutrition. **Named evidence/example:** Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Blood heart and double circulation. **Named evidence/example:** Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation

is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions.

Analysis: This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Respiration gas exchange and cellular energy. **Named evidence/example:** Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Kidney nephron and homeostasis. **Named evidence/example:** The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.
- Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions.
- Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous.
- The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys.

Qualified conclusion: Claim: Digestion absorption and nutrition. **Named evidence/example:** Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Blood heart and double circulation. **Named evidence/example:** Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions.

Analysis: This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Respiration gas exchange and cellular energy. **Named evidence/example:** Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Kidney nephron and homeostasis. **Named evidence/example:** The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **analyse** requires a direct position on "Analyse digestion, circulation, respiration and excretion as an integrated physiological...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Digestion absorption and nutrition. **Named evidence/example:** Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Blood heart and double circulation. **Named evidence/example:** Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Respiration gas exchange and cellular energy. **Named evidence/example:** Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Kidney nephron and homeostasis. **Named evidence/example:** The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions. **Analysis:** Connect mechanism ->

named system/institution -> application or implementation pathway -> public or strategic consequence.

Qualification: State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

3. **Claim and named evidence:** Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence.

Qualification: State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

4. **Claim and named evidence:** The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Digestion absorption and nutrition. **Named evidence/example:** Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Blood heart and double circulation. **Named evidence/example:** Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions.

Analysis: This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Respiration gas exchange and cellular energy. **Named evidence/example:** Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Kidney nephron and homeostasis. **Named evidence/example:** The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Analyse digestion, circulation, respiration and excretion as an integrated physiological...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Explain how nervous, endocrine and immune systems maintain regulated internal function. Answer in about 250 words.

Model thesis: Claim: Nervous endocrine and feedback control. **Named evidence/example:** Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Hormones metabolism and internal stability. **Named evidence/example:** Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Innate adaptive and tolerance architecture. **Named evidence/example:** Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems.
- Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms.
- Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone.

Qualified conclusion: Claim: Nervous endocrine and feedback control. **Named evidence/example:** Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Hormones metabolism and internal stability. **Named evidence/example:** Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value;

hormone deficiency, resistance and gland failure are different mechanisms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Innate adaptive and tolerance architecture. **Named evidence/example:** Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **explain** requires a direct position on "Explain how nervous, endocrine and immune systems maintain regulated internal function....", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Nervous endocrine and feedback control. **Named evidence/example:** Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Hormones metabolism and internal stability. **Named evidence/example:** Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Innate adaptive and tolerance architecture. **Named evidence/example:** Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit,

source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

3. **Claim and named evidence:** Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone.

Analysis: Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Nervous endocrine and feedback control. **Named evidence/example:** Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Hormones metabolism and internal stability. **Named evidence/example:** Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Innate adaptive and tolerance architecture. **Named evidence/example:** Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Explain how nervous, endocrine and immune systems maintain regulated internal function....", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Examine disease transmission, antimicrobial resistance, biofilms and microbiome interactions through a One Health and public-health lens. Answer in about 300 words.

Model thesis: **Claim:** Pathogen transmission vector and zoonosis map. **Named evidence/example:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Antibiotic antiviral and AMR boundary. **Named evidence/example:** Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biofilms microbiome probiotics and extremophiles. **Named evidence/example:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Health institution and evidence-status firewall. **Named evidence/example:** ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself.
- Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body.
- A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context.
- ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs.

Qualified conclusion: Claim: Pathogen transmission vector and zoonosis map. **Named evidence/example:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Antibiotic antiviral and AMR boundary. **Named evidence/example:** Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biofilms microbiome probiotics and extremophiles. **Named evidence/example:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Health institution and evidence-status firewall. **Named evidence/example:** ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **examine** requires a direct position on "Examine disease transmission, antimicrobial resistance, biofilms and microbiome interactions...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Pathogen transmission vector and zoonosis map. **Named evidence/example:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Antibiotic antiviral and AMR boundary. **Named evidence/example:** Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biofilms microbiome probiotics and extremophiles. **Named evidence/example:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Analysis:** This fixes the scientific

process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Health institution and evidence-status firewall. **Named**

evidence/example: ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: **Claim:** Pathogen transmission vector and zoonosis map. **Named evidence/example:**

Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Antibiotic antiviral and AMR boundary. **Named evidence/example:** Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body. **Analysis:** This

fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biofilms microbiome probiotics and extremophiles. **Named evidence/example:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Health institution and evidence-status firewall. **Named evidence/example:** ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Examine disease transmission, antimicrobial resistance, biofilms and microbiome interactions...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Discuss plant physiology, organism classification and cellulose decomposition while preserving biotechnology and vaccine owner boundaries. Answer in about 300 words.

Model thesis: Claim: Plant physiology transport and stress response. **Named evidence/example:** Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Routed taxonomy and organism-form distinctions. **Named evidence/example:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cellulose decomposition route. **Named evidence/example:** Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment

status boundary.

Claim -> named evidence -> analysis -> qualification:

- Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies.
- Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties.
- Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process.

Qualified conclusion: Claim: Plant physiology transport and stress response. **Named evidence/example:**

Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: Routed taxonomy and organism-form distinctions. **Named**

evidence/example: Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cellulose decomposition route. **Named evidence/example:** Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **discuss** requires a direct position on "Discuss plant physiology, organism classification and cellulose decomposition while...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Plant physiology transport and stress response. **Named evidence/example:**

Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Routed taxonomy and organism-form distinctions. **Named**

evidence/example: Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove

toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cellulose decomposition route. **Named evidence/example:** Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Plant physiology transport and stress response. **Named evidence/example:** Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Routed taxonomy and organism-form distinctions. **Named evidence/example:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cellulose

decomposition route. **Named evidence/example:** Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Discuss plant physiology, organism classification and cellulose decomposition while...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

Subject: Science & Technology | **Tier:** Advanced | **GS Paper:** GS-III + Prelims (*life-science fundamentals upgraded into public-health, biotechnology and governance analysis*). **Core area:** Gene regulation, human physiological systems, immunity, microbiology and plant-function basics as analytical tools for policy and technology questions. **Grounded in:** NCERT Class IX-X Science; NCERT Class XI-XII Biology; Nobel Prize in Physiology or Medicine 2024 press release (<https://www.nobelprize.org/prizes/medicine/2024/press-release/>); WHO trachoma validation page (<https://www.who.int/news/item/08-10-2024-elimination-of-trachoma-as-a-public-health-problem-in-india>); WHO India polio transition snapshot (<https://www.who.int/publications/m/item/india-polio-transition-snapshot>); WHO/UNICEF immunization-coverage methodology page (<https://www.who.int/teams/immunization-vaccines-and-biologicals/immunization-analysis-and-insights/global-monitoring/immunization-coverage/who-unicef-estimates-of-national-immunization-coverage>) - verified 16 Jul 2026. **FACT:** = source-grounded | **ANALYSIS:** = analytical inference | **CURRENT:** = current/dated development. **Companion:** basic/23_General-Science-Biology-and-Physiology.md.

1. Why this topic is analytically important

ANALYSIS: Biology in UPSC is not confined to naming organs or chromosomes. It is the conceptual foundation for public-health policy, diagnostics, biotechnology, nutrition, disease surveillance, genomics and even agricultural resilience. The advanced candidate turns cell-level processes into systems-level explanations.

2. Visual foundation

Biological layer	Core process	Public-system implication
FACT: Cell & organelles	Energy production, protein synthesis, transport	Disease, drugs, diagnostics
FACT: Genetics	Information storage, transmission and regulation	Heredity, genomics, biotechnology
FACT: Physiology	Homeostasis through organ-system coordination	Health, nutrition, disease burden
FACT: Immunology	Recognition, response, memory	Vaccination, surveillance, epidemic control
FACT: Microbiology	Pathogen type and host interaction	Infectious disease strategy, AMR, sanitation
FACT: Plant physiology	Photosynthesis, respiration, transport	Crop productivity, stress tolerance

DNA -> transcription -> RNA -> translation / regulation -> protein -> cell function -> tissue/organ function -> health outcome

3. Essential definitions

Concept	Precise meaning
FACT: Homeostasis	Maintenance of a relatively stable internal environment despite external change.
FACT: Gene expression	Process by which genetic information is used to produce functional products such as proteins or regulatory RNAs.
FACT: mRNA	Messenger RNA carrying coding information from DNA to ribosomes.
FACT: microRNA	Small regulatory RNA that modulates gene expression post-transcriptionally.
FACT: Reflex arc	Neural pathway mediating a rapid automatic response.
FACT: Endocrine gland	Ductless gland that secretes hormones into blood.
FACT: Pathogen	Disease-causing biological agent.
FACT: Vector	Organism that transmits pathogens between hosts.
FACT: Cell-mediated immunity	Adaptive immune response mainly involving T lymphocytes.
FACT: Humoral immunity	Adaptive immune response primarily mediated by antibodies in body fluids.

4. Mechanism / how it works

How gene information becomes physiology

1. FACT: DNA stores hereditary information, but that information influences life only when expressed.
2. FACT: Transcription produces RNA; different RNAs then help carry, regulate or translate this information.
3. FACT: Proteins produced through translation perform structural, catalytic, transport and signaling roles.
4. FACT: Cell behavior aggregates into tissue function, organ function and whole-body physiology.
5. FACT: Regulatory layers such as hormones, nerves and RNAs ensure that the same genome can produce very different cell types and responses.
6. ANALYSIS: This is why modern biology questions increasingly test regulation, not just storage of information.

5. Institutions and programmes

- FACT: **DBT**: biotechnology policy and programme institution linking basic biology to applied missions.
- FACT: **CSIR-CCMB / CSIR-IGIB**: molecular biology, genomics and systems-biology research ecosystems.
- FACT: **ICMR / disease-surveillance and public-health institutions**: where physiology, microbiology and immunology become policy action. ANALYSIS: Separate the functions precisely: **ICMR** performs and funds biomedical research; **NCDC and the IDSP/IHIP network under MoHFW** run operational surveillance and outbreak response; **State health departments** deliver programmes; **CDSCO** grants regulatory approvals. Research capacity, surveillance capacity and regulatory capacity fail in different ways and require different remedies.
- FACT: **WHO-linked surveillance and immunization reporting frameworks**: relevant because public-health biology is continuously monitored, not studied once and forgotten.
- ANALYSIS: Applied extensions sit in 13_Biotechnology-Fundamentals-and-DBT-Missions.md, 14_Genetic-Engineering-GM-Crops-and-CRISPR.md and 15_Vaccines-Monoclonal-Antibodies-and-Biopharma.md.

6. Indian applications, examples and limitations

- FACT: Genetics and cell biology underpin newborn screening, cancer diagnostics, pathogen surveillance and precision medicine.

- FACT: Physiology underlies nutrition programmes, anemia policy, non-communicable disease prevention and occupational health.
- FACT: Immunology shapes vaccination, antibody testing, epidemic preparedness and infection-control logic.
- FACT: Microbiology is central to sanitation, hospital infection control, water safety and antimicrobial-resistance management.
- FACT: Plant physiology matters for productivity, drought stress, nutrient-use efficiency and crop resilience.
- ANALYSIS: **Limitation 1:** biology-based policy often struggles because mechanism is known but delivery systems are weak.
- ANALYSIS: **Limitation 2:** public debate can become genetically deterministic, ignoring environment, nutrition and social conditions.
- ANALYSIS: **Limitation 3:** biology advances faster than public literacy, so misconceptions about genes, immunity or microbes become policy obstacles.

7. Deeper conceptual linkages

- ANALYSIS: **DNA vs RNA:** DNA is the long-term hereditary archive; RNA is not a mere copy-sheet but a functional and regulatory player. This distinction is mandatory before any later CRISPR discussion.
- ANALYSIS: **Nervous vs endocrine regulation:** one is rapid, wired and short-lived; the other is slower, chemical and often longer-acting. Many physiology questions turn on this contrast.
- ANALYSIS: **Innate vs adaptive immunity:** quick non-specific defense versus slower specific memory-capable defense - the backbone of public-health explanation.
- ANALYSIS: **Microbiology to governance:** pathogen type shapes intervention - antibiotics do not work against viruses, and sanitation has different roles across bacterial, viral and parasitic disease burdens.
- ANALYSIS: **Plant physiology to food security:** photosynthesis, respiration and water/nutrient transport set the ceiling for agricultural productivity and climate adaptation.

8. Must-Know Facts for Advanced Prelims

- FACT: Same DNA can produce different cell types because gene expression is regulated differently.
- FACT: RNA exists in multiple functionally important forms; it is not limited to messenger role alone.
- FACT: Endocrine control and neural control interact; they should not be treated as isolated silos.
- FACT: Antibody production belongs primarily to B-cell lineage, while many T-cell roles are regulatory or cytotoxic.
- FACT: Bacteria, viruses and fungi differ fundamentally in cellular organization and response logic.
- FACT: Photosynthesis stores solar energy in chemical form, whereas respiration releases usable energy from food molecules.
- FACT: Homeostasis is the integrating concept behind many physiology questions.

9. Advanced UPSC traps

- WRONG: **RNA is only an intermediate copy of DNA.** -> Regulatory RNAs such as microRNA show a deeper functional role.
- WRONG: **Immunity means antibodies only.** -> Cellular immunity, innate barriers and memory are equally important.
- WRONG: **Public-health success is only a medical issue.** -> WASH, surveillance, immunization and behavior change are biological-governance interactions.
- WRONG: **All microorganisms are harmful.** -> Many are beneficial or neutral; pathogenicity is context-specific.
- WRONG: **Plant physiology is only for agriculture-option students.** -> UPSC general science regularly uses it for food-security and environment-linked questions.

10. CURRENT: Current anchor -> analytical use

Verified current anchor	Topic-specific analytical use
CURRENT: 07 Oct 2024: Nobel Prize in Physiology or Medicine for microRNA and post-transcriptional gene regulation.	Use to show that RNA is functionally active in regulation, not merely a passive messenger.
CURRENT: 08 Oct 2024: WHO validated India's elimination of trachoma as a public-health problem.	Use to connect microbiology, hygiene, surveillance and long-term public-health delivery.
CURRENT: Apr 2024: WHO India polio transition snapshot.	Use to show that eradication success still requires surveillance and immunization-system maintenance.
CURRENT: Data reported till 21 Jun 2026: WHO/UNICEF annual immunization-estimate update cycle.	Use to connect immunology basics with real public-health monitoring systems.

Current as of 16 Jul 2026; verify for later updates.

12. Mains framework / angles

- ANALYSIS: Start with the basic mechanism: gene regulation, homeostasis, immune recognition or plant function.
- ANALYSIS: Then identify the policy interface: public health, nutrition, disease control, biotech or crop resilience.
- ANALYSIS: Respect the topic boundaries: CRISPR and applied gene editing go to Topic 14; vaccine platforms and regulation go to Topic 15.
- ANALYSIS: Use current anchors to show that fundamentals remain scientifically alive and institutionally relevant.

Answer thesis: Biology and physiology become GS-III material when molecular regulation, organ-system coordination and immunity are used to explain public health, biotechnology, disease surveillance and food-system resilience.

13. Probable questions

- ANALYSIS: **Practice Prelims:** Which of the following best captures the distinction between DNA, mRNA, microRNA and proteins?
- ANALYSIS: **Practice Mains (10 marks):** Why is the distinction between innate and adaptive immunity important for understanding public-health responses? *Answer in 150 words.*
- ANALYSIS: **Practice Mains (15 marks):** Discuss how basic genetics, physiology and microbiology together shape India's public-health and biotechnology priorities.

14. Study links

- FACT: Foundation companion: basic/23_General-Science-Biology-and-Physiology.md.
- FACT: 13_Biotechnology-Fundamentals-and-DBT-Missions.md - applied institutional and mission context.
- FACT: 14_Genetic-Engineering-GM-Crops-and-CRISPR.md - gene-editing and applied genetics extension.
- FACT: 15_Vaccines-Monoclonal-Antibodies-and-Biopharma.md - vaccine technology and regulation extension.
- FACT: 22_General-Science-Chemistry-Fundamentals.md - biomolecules from chemistry to life-science function.

CONSOLIDATED REGISTER NOTES

General Science: Biology and Physiology: CELL, BIOMOLECULE, GENETICS AND MEMBRANE RAPID MAP

1. **Cell organisation and organelle boundary:** A cell is the basic structural and functional unit of life; prokaryotes lack a membrane-bound nucleus and membrane-bound organelles, whereas eukaryotes possess them. The nucleus stores most genetic material, ribosomes synthesize protein, mitochondria support aerobic ATP generation, the endoplasmic reticulum and Golgi system process and move products, lysosomes contain hydrolytic enzymes, and plant cells additionally possess a cellulose wall, chloroplasts and a large vacuole.
2. **Biomolecules and enzyme specificity:** Carbohydrates commonly supply or store energy and provide structures such as cellulose, lipids store energy and form membranes, proteins perform structural, transport, receptor and catalytic roles, and nucleic acids store or express information. Enzymes are biological catalysts whose activity depends on substrate fit and conditions such as temperature and pH; a biomolecule class does not establish one universal function.
3. **Membrane transport and water balance:** Diffusion moves particles down a concentration gradient, osmosis is net water movement across a selectively permeable membrane, and active transport uses cellular energy to move substances against a gradient. Tonicity describes the effect of an external solution on cell water balance, so swelling, plasmolysis and normal volume require the membrane and concentration context.
4. **Mitosis meiosis and variation:** Mitosis is one division producing two genetically similar daughter cells for growth and repair, while meiosis consists of two divisions producing four haploid products for sexual reproduction; crossing over and independent assortment generate variation. Neither process should be described as gene editing, and chromosome number must be tracked relative to the starting cell.
5. **DNA RNA gene expression and microRNA:** DNA replication copies hereditary material, transcription produces RNA from a DNA template, and translation uses messenger RNA, transfer RNA and ribosomes to build protein from codons. RNA also has regulatory roles: microRNA can bind complementary messenger RNA and reduce protein production or promote RNA degradation, so DNA storage, RNA regulation and protein function are distinct stages.
6. **Inheritance mutation and sex linkage:** A gene is a functional hereditary unit on DNA; alleles are alternative forms, segregation separates allele pairs into gametes and independent assortment applies to appropriately unlinked genes. Mutations may be point, frameshift or chromosomal changes and supply variation without guaranteeing benefit; human XX-XY sex determination and X-linked recessive inheritance explain why haemophilia or red-green colour blindness can show sex-biased patterns.
7. **Digestion absorption and nutrition:** Digestion converts food into absorbable molecules: salivary amylase begins starch digestion, pepsin works in acidic stomach conditions, bile emulsifies fat without acting as an enzyme, pancreatic and intestinal enzymes continue digestion, and the small intestine is the principal absorption site. Vitamins, minerals, protein and energy have different roles; deficiency, absorption failure and excess are separate nutritional problems.
8. **Blood heart and double circulation:** Blood contains plasma, red cells carrying haemoglobin, immune white cells and platelets involved in clotting; the four-chambered human heart supports pulmonary and systemic circuits. Arteries carry blood away from the heart and veins return it, so oxygenation is not the defining rule; ABO and Rh compatibility, gas transport and clotting must be treated as different functions.
9. **Respiration gas exchange and cellular energy:** Ventilation moves air, alveolar gas exchange moves oxygen and carbon dioxide across thin surfaces, blood transports gases, and cellular respiration releases usable energy from nutrients. Haemoglobin binds oxygen while carbon monoxide can disrupt transport through stronger binding; breathing, gas exchange and ATP-producing respiration are linked but not synonymous.
10. **Kidney nephron and homeostasis:** The nephron filters blood at the glomerulus and then selectively reabsorbs and secretes substances along its tubule to regulate water, electrolytes, acid-base balance and wastes. Dialysis can replace selected filtration functions when kidneys fail but does not reproduce every endocrine and regulatory role of healthy kidneys.
11. **Nervous endocrine and feedback control:** Neurons transmit rapid electrical signals and communicate chemically at synapses, reflex arcs permit fast automatic responses, and the brain and spinal cord coordinate

wider activity. Endocrine glands release hormones into blood for target-dependent effects that are often slower and longer-lasting; neural and hormonal control interact through feedback rather than operating as isolated systems.

12. **Hormones metabolism and internal stability:** Insulin lowers elevated blood glucose by promoting uptake and storage while glucagon supports its rise, thyroid hormones influence metabolism, parathyroid hormone helps regulate calcium, adrenal hormones mediate stress responses and pituitary signals coordinate several glands. Homeostasis is dynamic regulation around a range, not an unchanging value; hormone deficiency, resistance and gland failure are different mechanisms.
13. **Innate adaptive and tolerance architecture:** Innate immunity includes barriers, phagocytes, natural killer cells, complement, interferons, inflammation and fever; adaptive immunity uses antigen-specific B and T lymphocytes and can form memory. B-cell lineages produce antibodies, T cells perform coordinating, cytotoxic or regulatory roles, and peripheral immune tolerance including regulatory T cells limits attack on self; immunity is not antibodies alone.
14. **Pathogen transmission vector and zoonosis map:** Bacteria are cellular organisms, viruses are acellular and depend on host-cell machinery, fungi are eukaryotes, protozoa are single-celled eukaryotes and helminths are multicellular parasites. Transmission may be respiratory, faeco-oral, blood-borne, sexual, vector-borne, fomite-linked or zoonotic; a vector such as Aedes, Anopheles, Culex or sandfly carries a pathogen but is not the disease itself.
15. **Antibiotic antiviral and AMR boundary:** Antibiotics target susceptible bacteria rather than viruses, while antivirals interfere with virus-specific replication stages. Antimicrobial resistance is selection and spread of resistant microorganisms under pressures including misuse, incomplete treatment, agricultural use and environmental residues; the microbe or population becomes resistant, not the patient's body.
16. **Biofilms microbiome probiotics and extremophiles:** A biofilm is a surface-associated microbial community embedded in a matrix that can alter persistence and treatment response; the human microbiome contributes to digestion, vitamin production, immune development and pathogen exclusion. Probiotics are live microorganisms intended to confer a benefit in an adequate amount, while thermophiles, psychrophiles and acidophiles are classified by environmental tolerance; none of these labels proves safety or effectiveness in every context.
17. **Plant physiology transport and stress response:** Photosynthesis stores light energy in chemical form while plants also respire continuously; xylem conducts water and minerals and phloem translocates organic food. Stomata regulate gas exchange and water loss, transpiration contributes to upward water movement, and auxin, gibberellin, cytokinin, abscisic acid and ethylene have distinct growth or stress roles; C3, C4 and CAM are different carbon-fixation strategies.
18. **Routed taxonomy and organism-form distinctions:** Groundnut, horse-gram and soybean are routed pea-family examples; cicada, frog hopper and pond skater are insects; poisonous species occur in multiple animal groups, so a broad group name does not prove toxicity. Cassava is a woody shrub, ginger a rhizomatous herb, Malabar spinach a climbing vine, mint a herb and papaya a large herb; mushrooms can have species-specific edible, medicinal, psychoactive, insect-pathogenic or bioluminescent properties.
19. **Cellulose decomposition route:** Cellulose is a structural plant polysaccharide; decomposer fungi and bacteria secrete cellulases that hydrolyse it into smaller sugars, after which microbial metabolism returns carbon through soil and food-web processes. Moisture, temperature, oxygen, lignin and community composition constrain the rate, and natural decomposition is not proof of an industrially viable biofuel process.
20. **Health institution and evidence-status firewall:** ICMR performs and supports biomedical research, NCDC and the IDSP or IHIP network support surveillance and outbreak response under MoHFW, State systems deliver programmes, and CDSCO regulates drugs and vaccines; DBT, CSIR-CCMB and CSIR-IGIB occupy different research roles. A biological mechanism, association, surveillance signal, WHO validation, elimination as a public-health problem, regulatory approval and eradication are separate evidence rungs.

General Science: Biology and Physiology: HUMAN PHYSIOLOGY, HOMEOSTASIS AND IMMUNITY SYSTEM GRID

- Do not merge prokaryotic and eukaryotic cells or assign every organelle to every cell.
- Do not give every carbohydrate, lipid, protein or enzyme one universal function.
- Do not confuse diffusion, osmosis and energy-dependent active transport.
- Do not merge mitosis with meiosis or crossing over with gene editing.
- Do not reduce RNA to a passive messenger or equate transcription with translation.
- Do not treat a mutation, gene association or sex-linked pattern as deterministic destiny.
- Do not call bile an enzyme or the stomach the principal site of nutrient absorption.
- Do not define arteries and veins by oxygen content.
- Do not use breathing, alveolar gas exchange and cellular respiration as synonyms.
- Do not claim dialysis reproduces every kidney function.
- Do not merge a neural impulse, neurotransmitter, enzyme and hormone.
- Do not describe homeostasis as a perfectly fixed internal value.
- Do not equate immunity with antibodies alone or an antigen with an antibody.
- Do not treat bacteria, viruses, fungi, protozoa, helminths and vectors as one category.
- Do not route vaccine platforms, monoclonal antibodies or gene-editing mechanisms into this fundamentals owner.

General Science: Biology and Physiology: DISEASE, MICROBE, PLANT AND ROUTED-PYQ FIREWALLS

IDENTIFY CELL TYPE ORGANELLE BIOMOLECULE MEMBRANE AND GRADIENT
-> TRACE DNA REPLICATION TRANSCRIPTION RNA REGULATION TRANSLATION AND PROTEIN
-> SEPARATE MITOSIS MEIOSIS MUTATION VARIATION AND GENE EDITING
-> FOLLOW DIGESTION ABSORPTION CIRCULATION GAS EXCHANGE RESPIRATION AND EXCRETION
-> MAP NEURAL SIGNAL HORMONE TARGET FEEDBACK AND HOMEOSTATIC RANGE
-> CLASSIFY INNATE ADAPTIVE HUMORAL CELLULAR MEMORY AND TOLERANCE
-> NAME PATHOGEN RESERVOIR TRANSMISSION ROUTE VECTOR AND PREVENTION LAYER
-> EXPLAIN AMR AS MICROBIAL SELECTION AND BIOFILM AS A COMMUNITY STATE
-> CONNECT PHOTOSYNTHESIS XYLEM PHLOEM STOMATA HORMONES AND STRESS STRATEGY
-> ROUTE TAXONOMY PLANT FORM NITRIC OXIDE VIRUS AND CELLULOSE PYQS WITHOUT KEYS
-> KEEP ICMR NCDC STATE DELIVERY CDSO DBT AND CSIR ROLES DISTINCT
-> STOP AT MECHANISM ASSOCIATION SURVEILLANCE VALIDATION APPROVAL OR OUTCOME

General Science: Biology and Physiology: PUBLIC-HEALTH INSTITUTION, STATUS AND ANSWER-WRITING SPINE

Official-source checks on 2026-09-04 retained the 2024 microRNA citation, the 2025 peripheral-immune-tolerance citation, WHO's trachoma validation and polio-transition boundaries, and WHO/UNICEF's annual estimate methodology. No therapy approval, vaccine-platform claim, coverage percentage, eradication claim, medical advice or PYQ key was invented.

COMPLETE TOPIC ASCII MASTER FLOW DIAGRAM

ASCII MASTER FLOW - PANEL 1/12: Cell and organelle map

PROKARYOTE -> no membrane-bound nucleus or organelles
EUKARYOTE -> nucleus + compartmental organelles
RIBOSOME -> protein | MITOCHONDRION -> ATP
ER + GOLGI -> processing and movement
PLANT -> cellulose wall + chloroplast + large vacuole

ASCII MASTER FLOW - PANEL 2/12: Membrane transport decision tree

SOLUTE DOWN GRADIENT -> diffusion
WATER + SELECTIVE MEMBRANE -> osmosis
AGAINST GRADIENT + ENERGY -> active transport
EXTERNAL TONICITY -> cell water response
TRAP -> movement depends on membrane and gradient

ASCII MASTER FLOW - PANEL 3/12: Cell division comparison

MITOSIS -> one division -> two similar cells
ROLE -> growth and repair
MEIOSIS -> two divisions -> four haploid products
ROLE -> gametes + variation
CROSSING OVER != gene editing

ASCII MASTER FLOW - PANEL 4/12: Gene-expression rail

DNA --replication--> DNA
DNA --transcription--> mRNA
mRNA + tRNA + RIBOSOME --translation--> protein
microRNA -> post-transcriptional regulation
PROTEIN -> cell structure / enzyme / receptor / signal

ASCII MASTER FLOW - PANEL 5/12: Nutrition and digestion route

MOUTH -> amylase
STOMACH -> acid + pepsin
LIVER / BILE -> fat emulsification
PANCREAS + INTESTINE -> digestion completion
SMALL INTESTINE -> principal absorption

ASCII MASTER FLOW - PANEL 6/12: Transport respiration excretion loop

LUNGS -> alveolar gas exchange
BLOOD -> gases + nutrients + hormones + wastes
HEART -> pulmonary + systemic circulation
CELLS -> ATP-producing respiration
NEPHRON -> filtration + reabsorption + secretion

ASCII MASTER FLOW - PANEL 7/12: Control and feedback ladder

RECEPTOR -> change detected
NERVOUS -> rapid wired response
ENDOCRINE -> blood-borne hormone response
TARGET -> physiological adjustment
NEGATIVE FEEDBACK -> restores regulated range

ASCII MASTER FLOW - PANEL 8/12: Immunity architecture

BARRIERS + PHAGOCYTES + COMPLEMENT -> innate
B CELL -> antibody / memory
T CELL -> helper / cytotoxic / regulatory roles
REGULATORY T CELL + Foxp3 -> peripheral tolerance
SELF-REACTIVITY CONTROL != no immune response

ASCII MASTER FLOW - PANEL 9/12: Disease transmission grid

BACTERIUM | VIRUS | FUNGUS | PROTOZOAN | HELMINTH
AIR | WATER/FOOD | BLOOD | CONTACT | VECTOR
VECTOR -> carries pathogen, is not disease
ZOOONOSIS -> animal-human interface
PREVENTION -> mechanism-specific layers

ASCII MASTER FLOW - PANEL 10/12: Microbial ecology and resistance

ANTIMICROBIAL PRESSURE -> selection
RESISTANT MICROBE -> survival + spread
BIOFILM -> matrix-protected community
MICROBIOME / PROBIOTIC -> context-dependent roles
EXTREMOPHILE LABEL -> tolerance, not universal safety
MUST REMEMBER: Biology and physiology connect cell structure, metabolism, genetics,...
MUST REMEMBER: Biology and physiology connect cell structure, metabolism, genetics,...

ASCII MASTER FLOW - PANEL 11/12: Plant function flow

ROOT -> water + mineral uptake
XYLEM -> upward transport
LEAF -> stomata + photosynthesis + transpiration
PHLOEM -> food translocation
C3 | C4 | CAM -> different carbon-fixation strategies
CLOSE DISTINCTION: Gene is not allele or genome, genotype is not phenotype, pathogen is...
CLOSE DISTINCTION: Gene is not allele or genome, genotype is not phenotype, pathogen is...

ASCII MASTER FLOW - PANEL 12/12: PYQ and evidence-status rail

TAXON / PLANT FORM / SPECIES PROPERTY -> exact classification
CELLULOSE -> cellulase -> sugars -> microbial metabolism
RESEARCH -> SURVEILLANCE -> PROGRAMME DELIVERY -> REGULATION
WHO VALIDATION != pathogen eradication
MECHANISM / ASSOCIATION / APPROVAL / OUTCOME -> separate rungs
EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name organism/cell/tissue/organ,...
EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name organism/cell/tissue/organ,...